DartCraft

Setup Guide

Connect • Calibrate • Play

Thank you for choosing DartCraft.

This guide covers everything you need to set up your LED ring, connect your cameras, and get started with Autodarts.

Contents

1 Dartboard Height & Throw Line

Before mounting your dartboard, ensure it is positioned at the correct regulation height. These measurements apply to standard steel-tip dartboards.

Measurement	Value
Bullseye height	**173 cm from the floor**
Throw distance	**237 cm from face of the board**

⚠️ *The bullseye must be exactly 173 cm from the floor. The throw line (oche) must be at least 237 cm from the face of the board — measure from the front surface of the board, not the wall.*

2 LED Ring Setup

The Autodarts ring provides integrated LED lighting and holds the three cameras. The following steps cover full ring assembly, LED strip installation, and camera leg assembly.

2.1 Ring Assembly

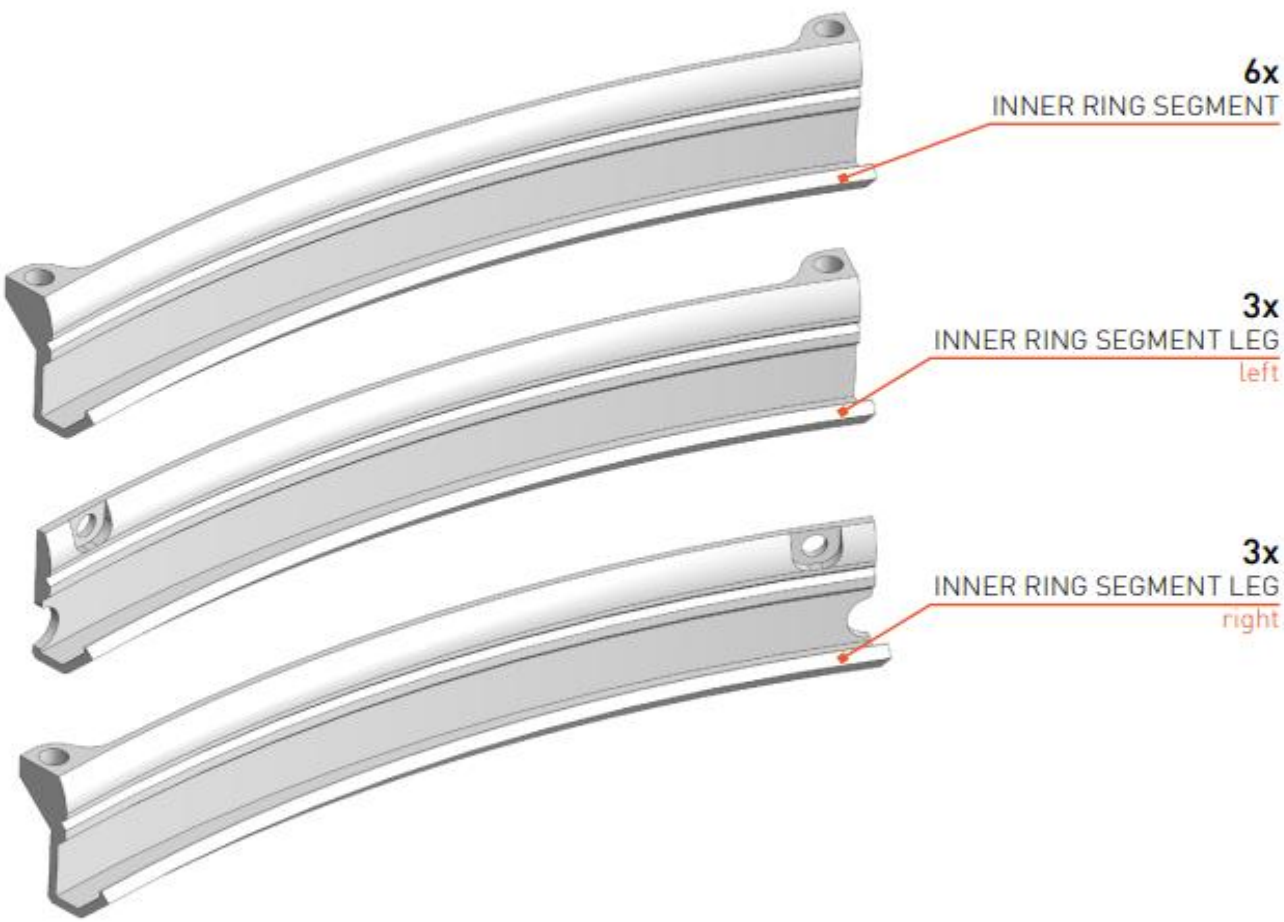
The ring is made up of 9 outer segments and a set of inner ring segments. Follow the steps below to assemble the full ring and fit the inner ring.

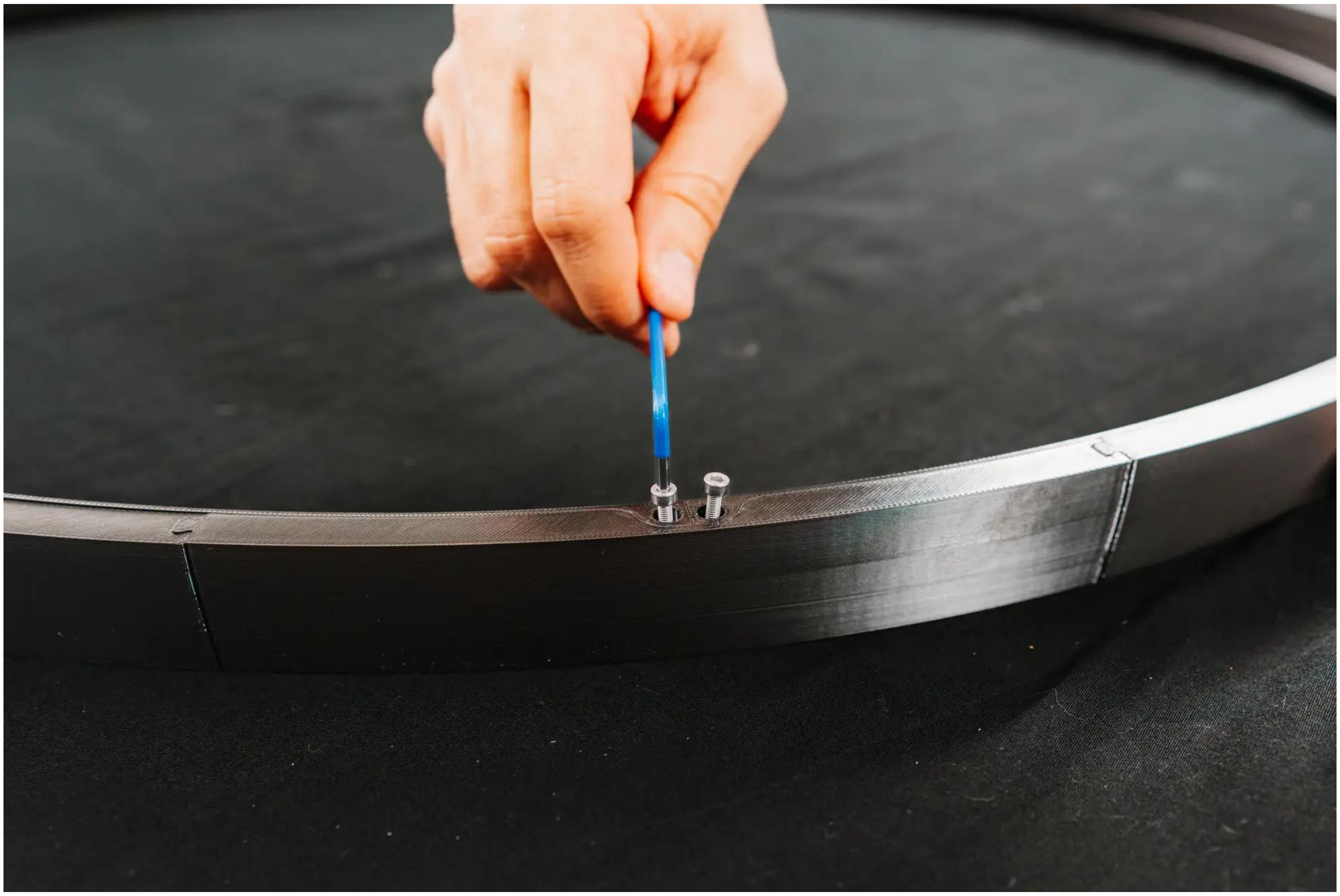
1. Gather the 9 main ring segments (see photo below). Connect them firmly together in groups, pressing each joint fully until seated, to form 3 separate sub-sections.

2. Using the 3 leg-mount segments, connect one between each sub-section to complete the full ring. Ensure all joints are fully clicked into place before proceeding.

3. Place one inner ring segment at each joint around the inside of the main ring. The inner ring strengthens the overall structure and prevents light from the LEDs bleeding through the outer joints.

   At the three positions where the camera legs will be mounted, use the inner ring leg segments with a hole. Once all the inner ring pieces are in place, secure each with the included M3 screws.



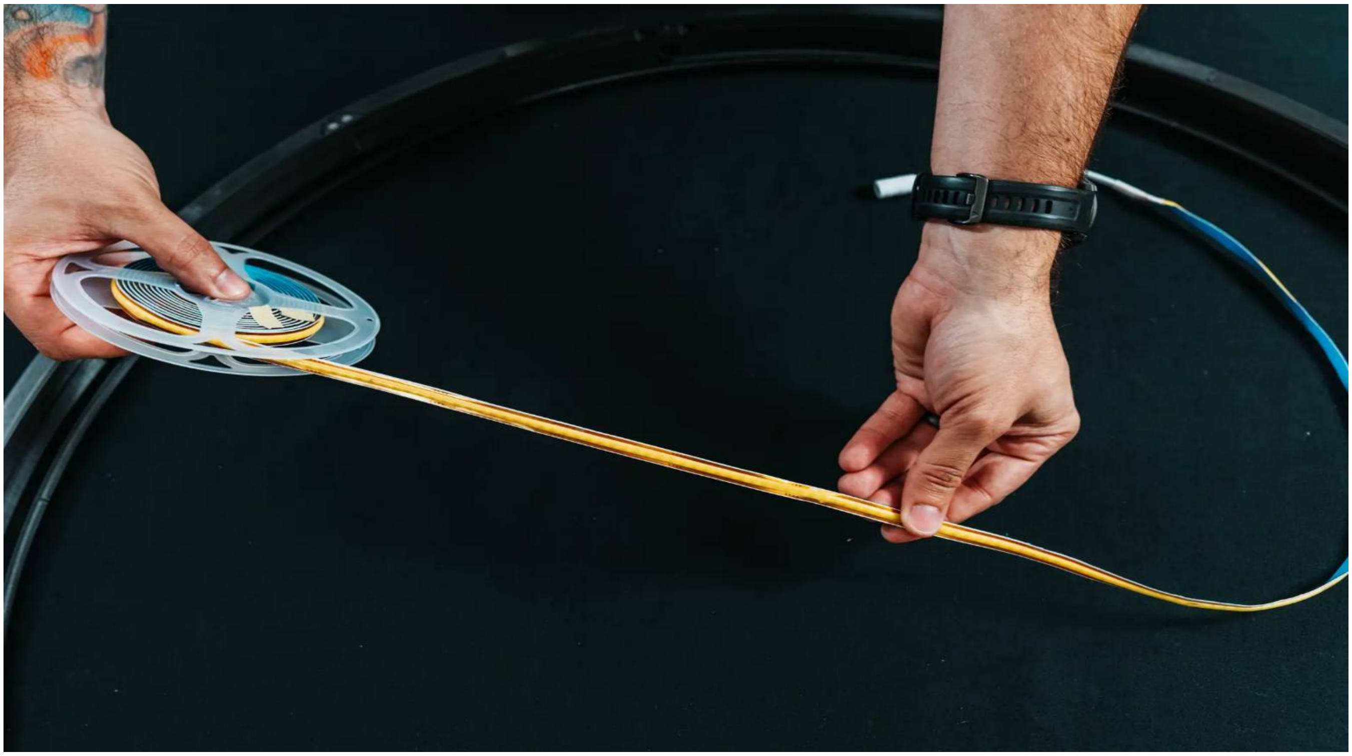


4. Once all the inner ring pieces are in place, secure each with the included M3 screws.

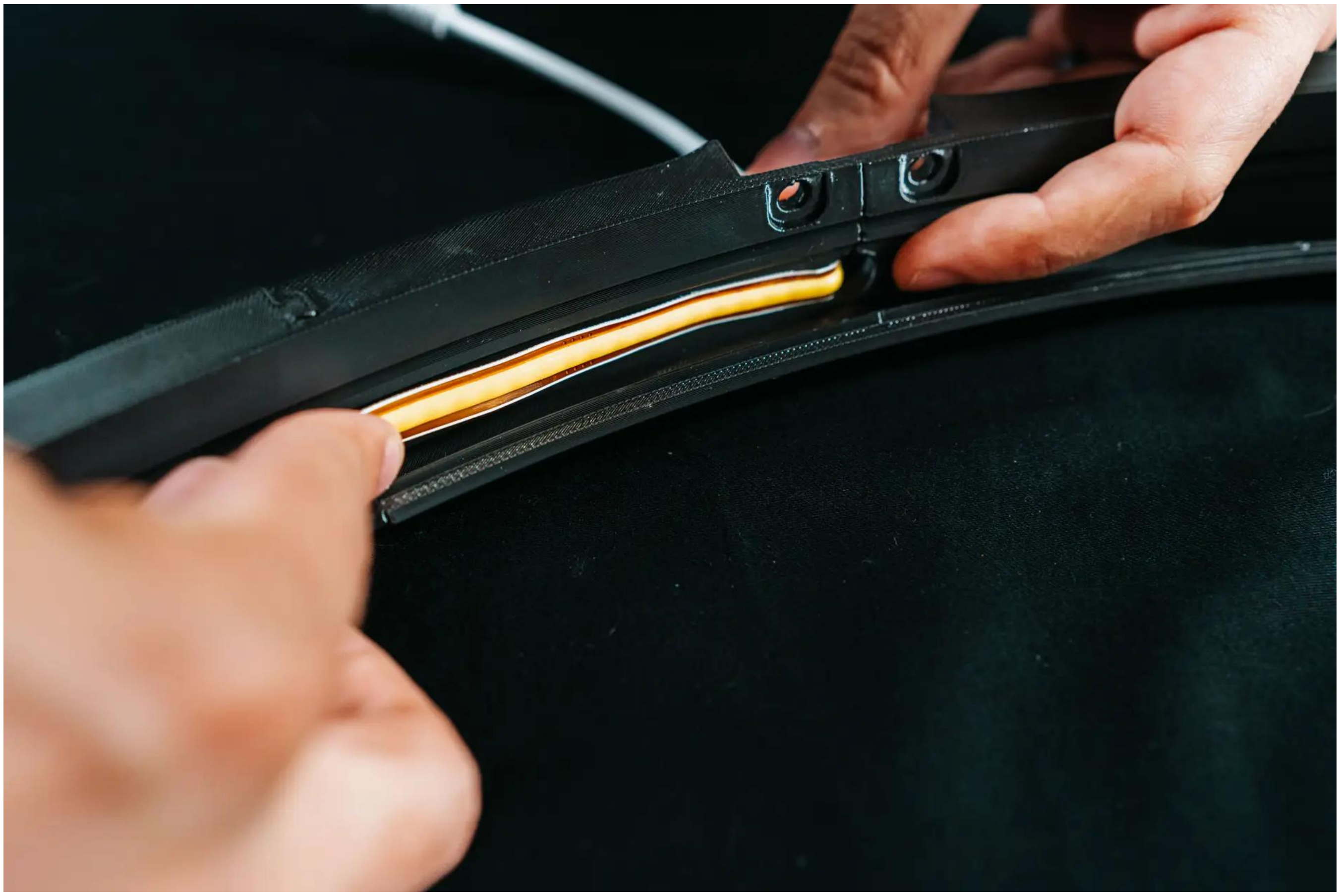
2.2 LED Strip Installation

The LED strip runs the full circumference of the ring and is routed through one of the leg-mount pieces to bring the power connector to the outside.

1. Wipe down the inner wall of the ring with isopropyl alcohol and allow it to dry completely. This ensures a strong, lasting bond.
2. Fully unwind the LED strip from its holder. From the outside of the ring, feed the strip through the larger hole in one of the leg-mount pieces, leaving just the power connector visible on the outside.



3. Starting at the end closest to the power connector, peel back a short section of the adhesive backing and press the strip firmly into the inner wall groove. Continue peeling and pressing, a section at a time, working your way around the ring until the strip is fully installed.



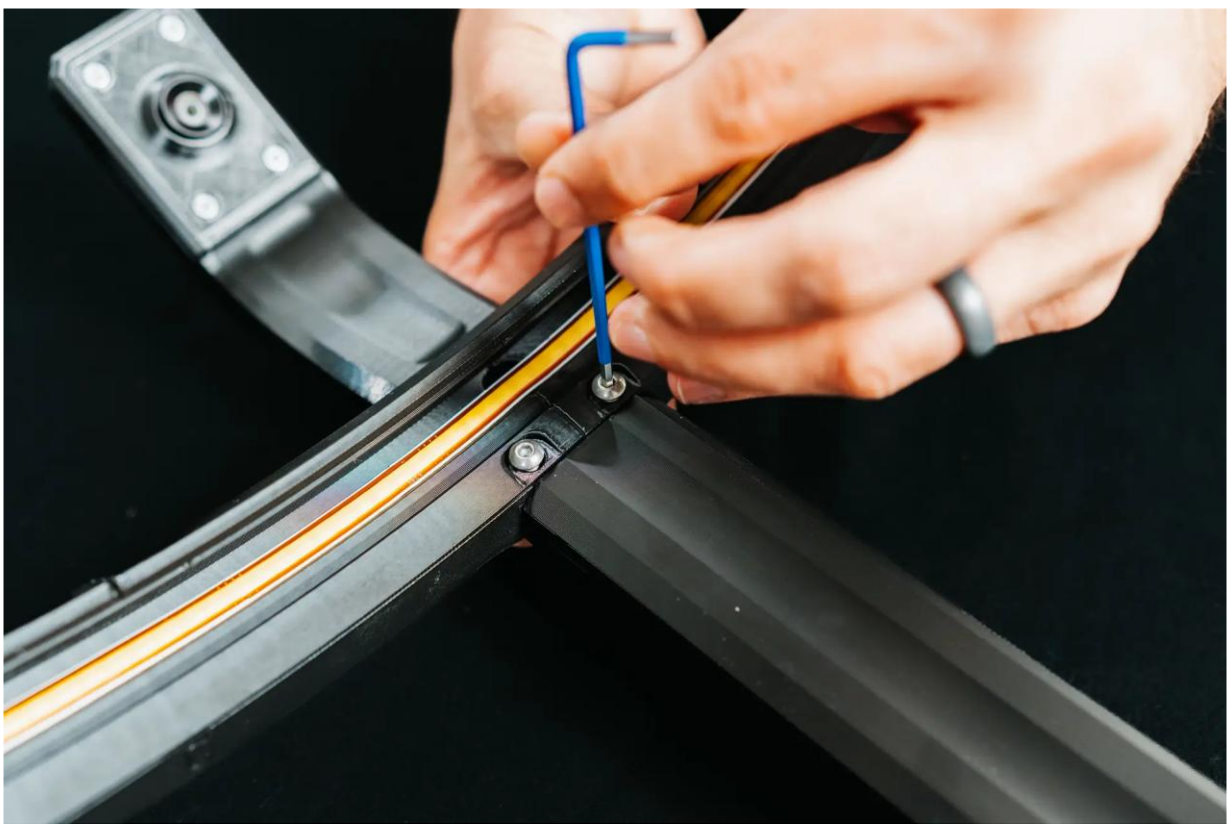
Tip: For extra bonding strength, reinforce the strip with 3M double-sided tape (1 cm width) along the groove before application.

2.3 Ring Legs Assembly

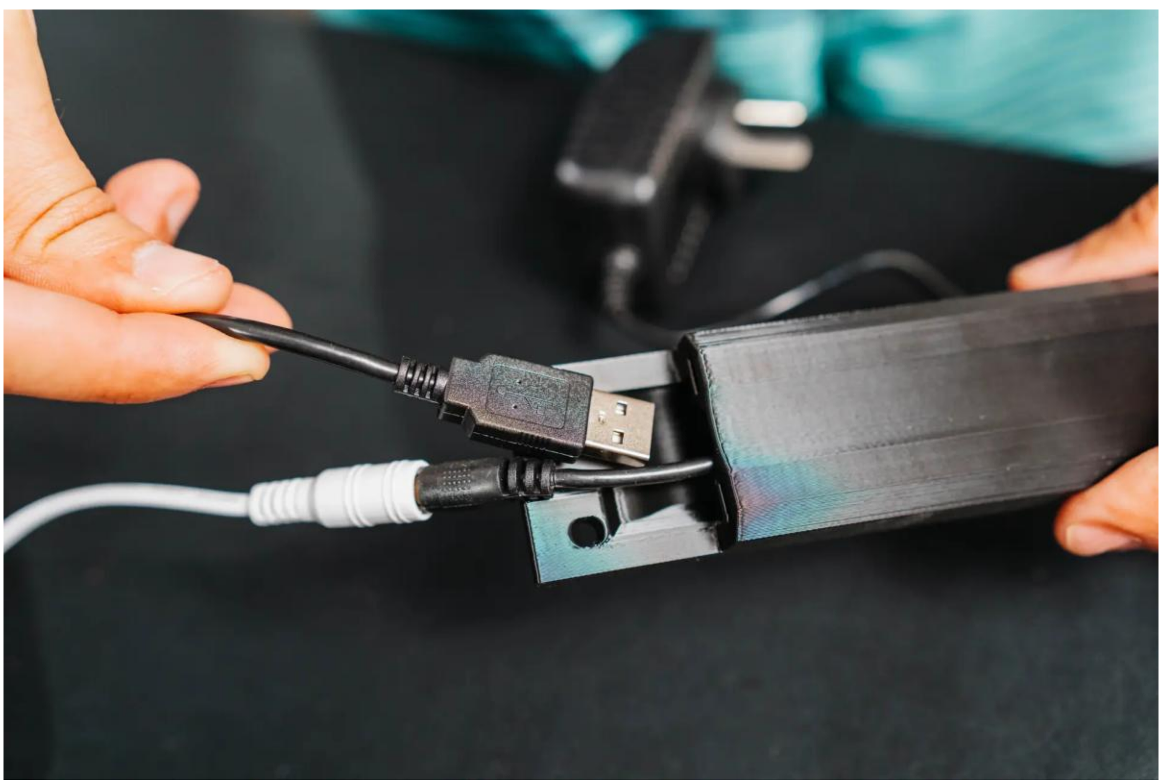
Each leg consists of an upper section with the camera already pre-installed and a lower section. Assemble the first two standard legs before moving on to the LED power leg.

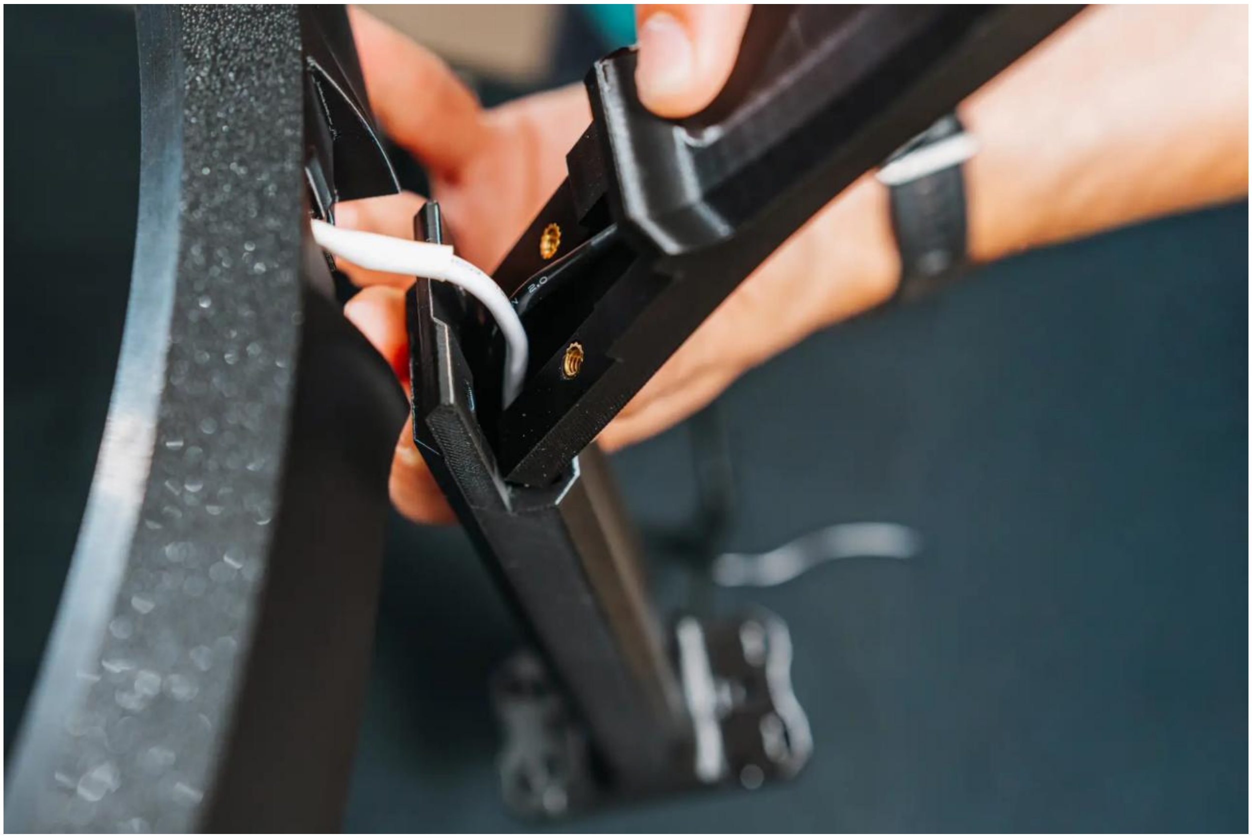
1. For each of the first two camera legs: feed the camera USB cable down through the lower leg, then insert the two hinge pins on the upper leg into the lower leg and press firmly together until secure. Repeat for the second leg.

2. Attach the two assembled camera legs to the ring using the included M4 screws, tightening them securely.



3. For the third leg at the LED strip power connector: feed the LED power supply cable through the lower leg first, then feed the camera USB cable through in the same way. Press the upper and lower leg together. Connect the LED strip connector to the LED power supply cable, then attach this leg to the ring using the included M4 screws.



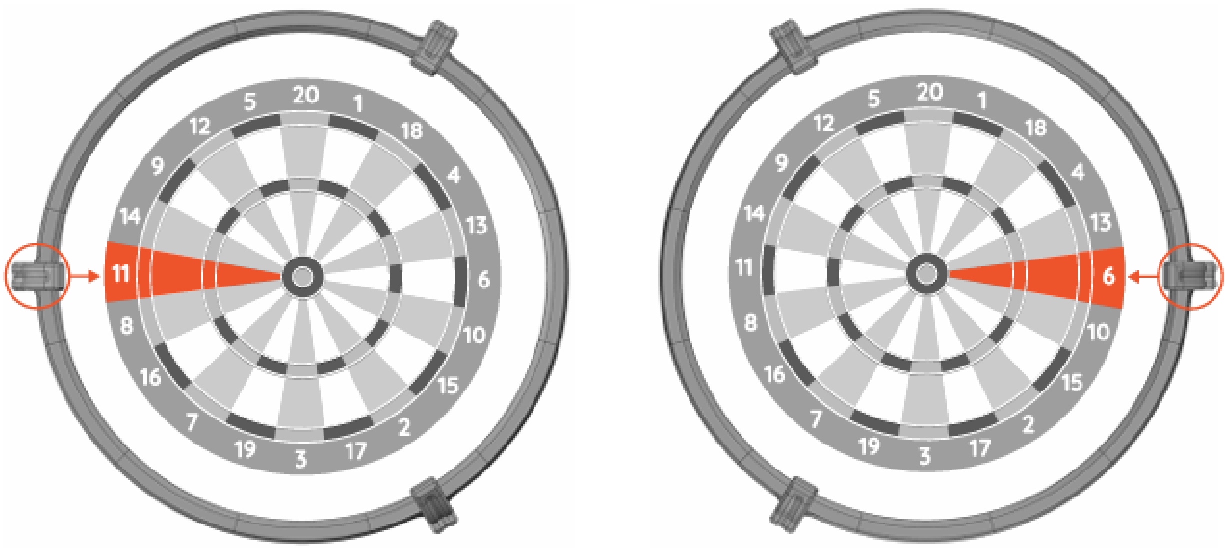


2.4 Camera Positioning

There are two valid camera positioning options for Autodarts. Choose one and apply it consistently — your camera legs must align with either dartboard segment 6 or segment 11 on the horizontal axis.

AUTODARTS CAMERA POSITIONING

You have two options where must be placed cameras around the dartboard.



Camera positioning tip no. 1
Camera alligned with dartboard segment 11

Camera positioning tip no. 2
Camera alligned with dartboard segment 6

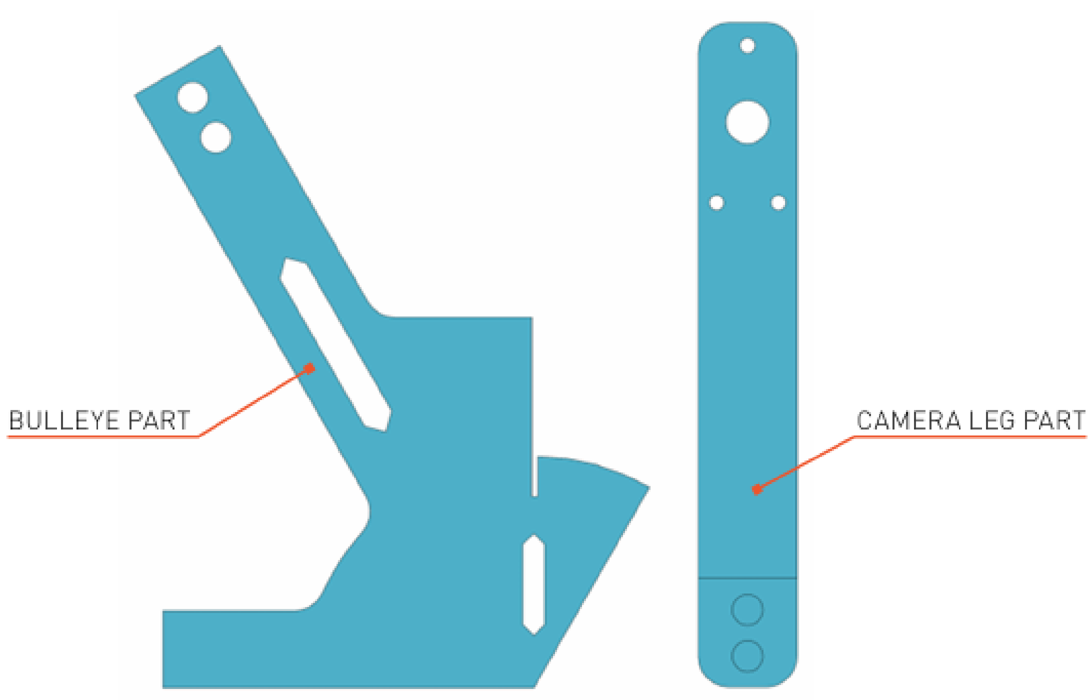
2.5 Wall Installation

Use the supplied two-part installation template to accurately mark the three leg mounting hole positions on the wall. The template must be positioned from the exact bullseye centre — remove the dartboard first if already mounted.

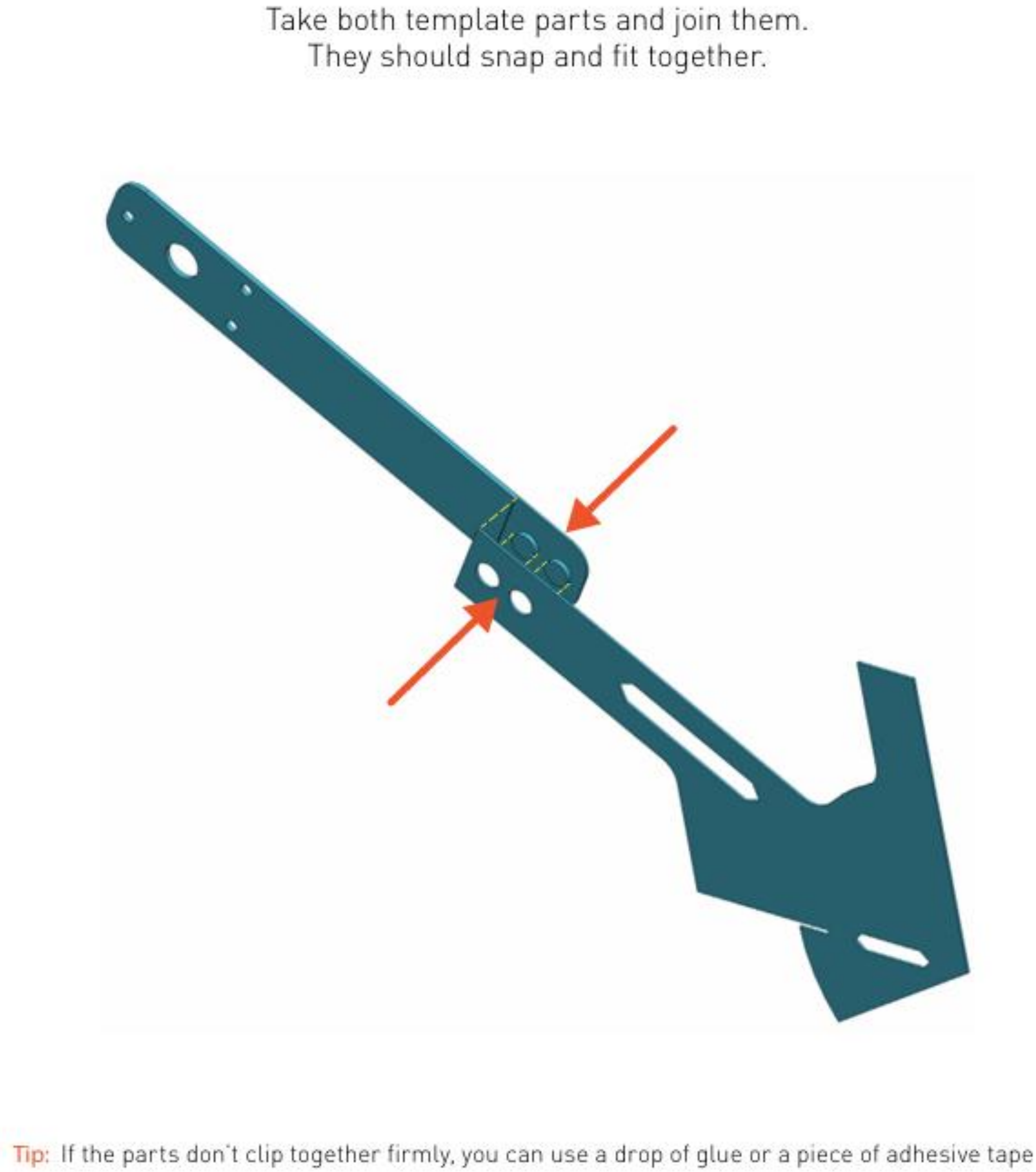
WALL INSTALLATION TIPS

To make it easier to mount the entire Autodarts ring centred on the dartboard, we have prepared a template that lets you measure the three leg positions.

The only requirement is that the template is positioned from the exact centre "bullseye", so it is necessary to remove the dartboard if it is already mounted on the wall.



Template parts: Bullseye part (left) and Camera Leg part (right)



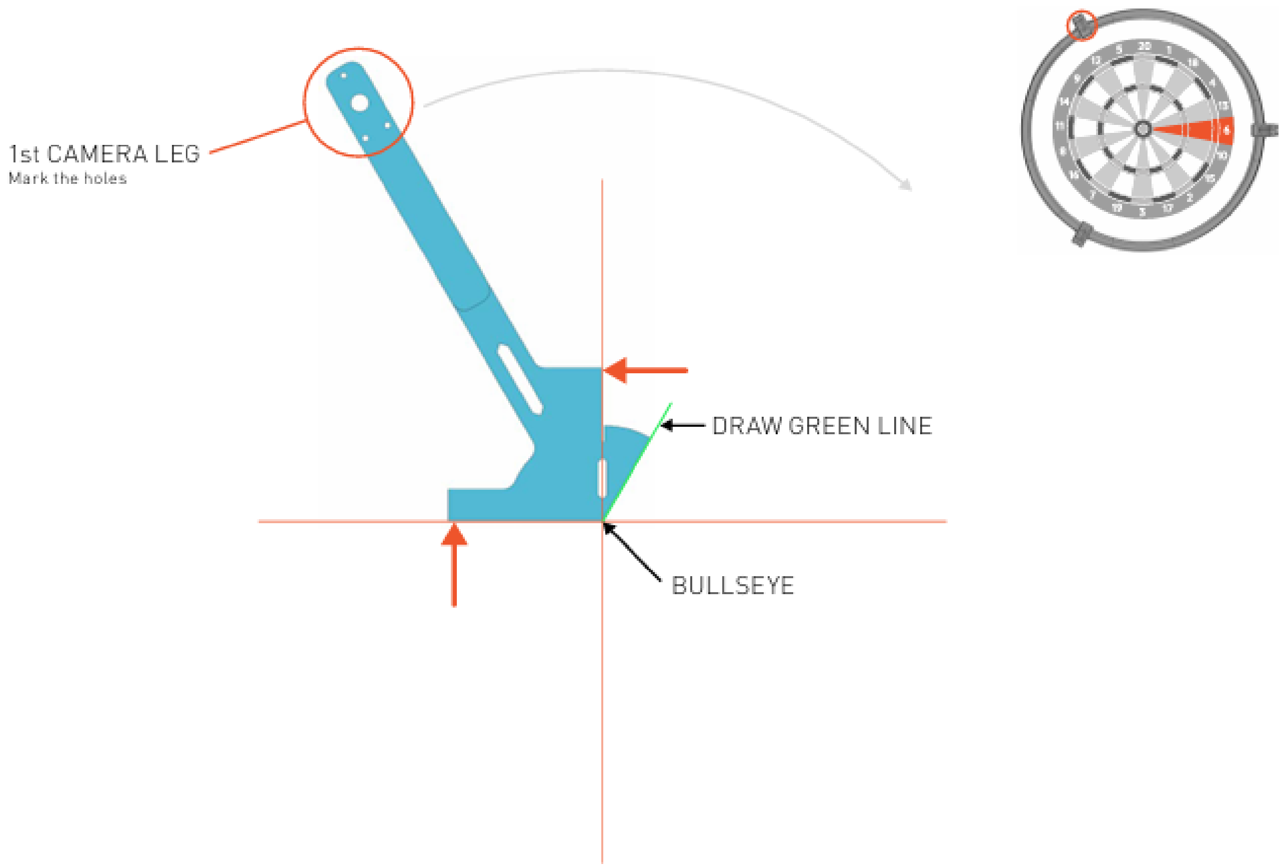
Join the two template parts — they should snap together. Use a drop of glue or tape if needed.

Option A — Camera on Segment 6

Step 1: Draw a vertical and horizontal line through the bullseye using a spirit level. Place the template corner at the bullseye centre, aligned with those lines. Mark the 1st camera leg holes and draw a green reference line.

USING A TEMPLATE - CAMERA ON SEGMENT no.6

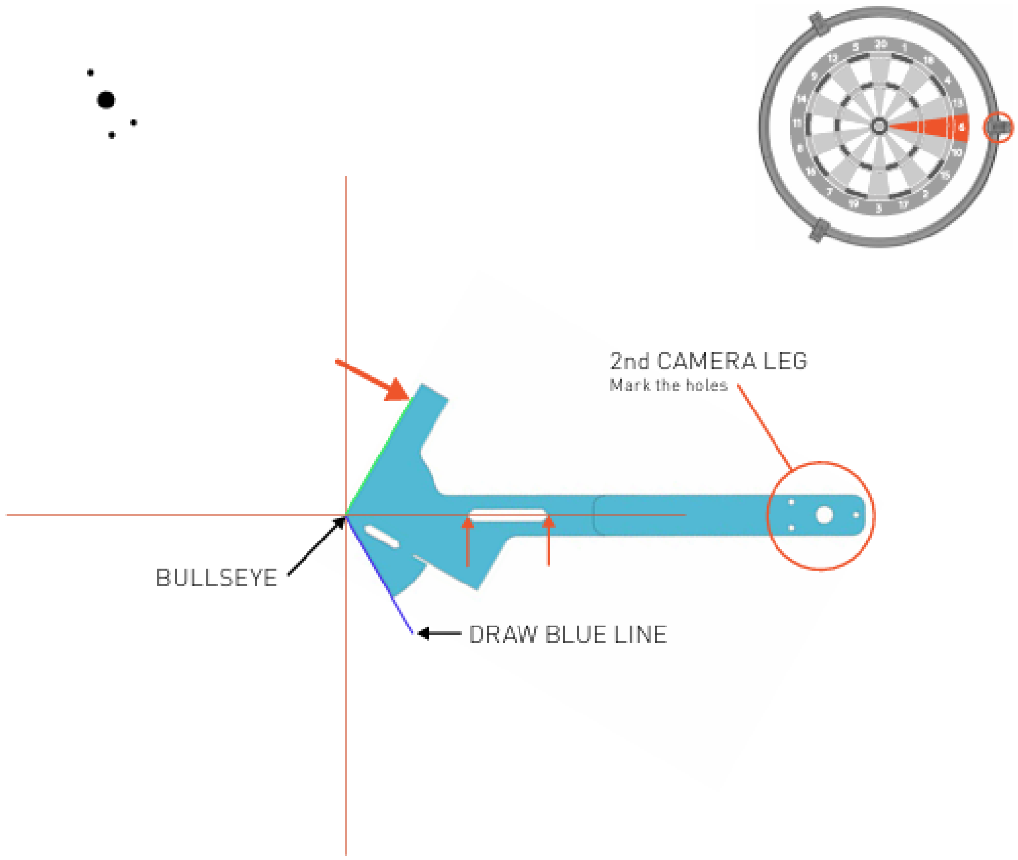
It is necessary to draw a vertical and horizontal line through the "bullseye". Please use a spirit level. Place the corner of the template on the centre of the "bullseye", and align it with the vertical and horizontal lines.
Mark the location of the second camera leg and draw a green line.



Step 1 — Mark 1st leg and draw green line

Rotate the model clockwise, and align it to the centre and the newly drawn green line.
You can also check the horizontal direction through the template hole,
the horizontal line should pass through the corners of the hole.

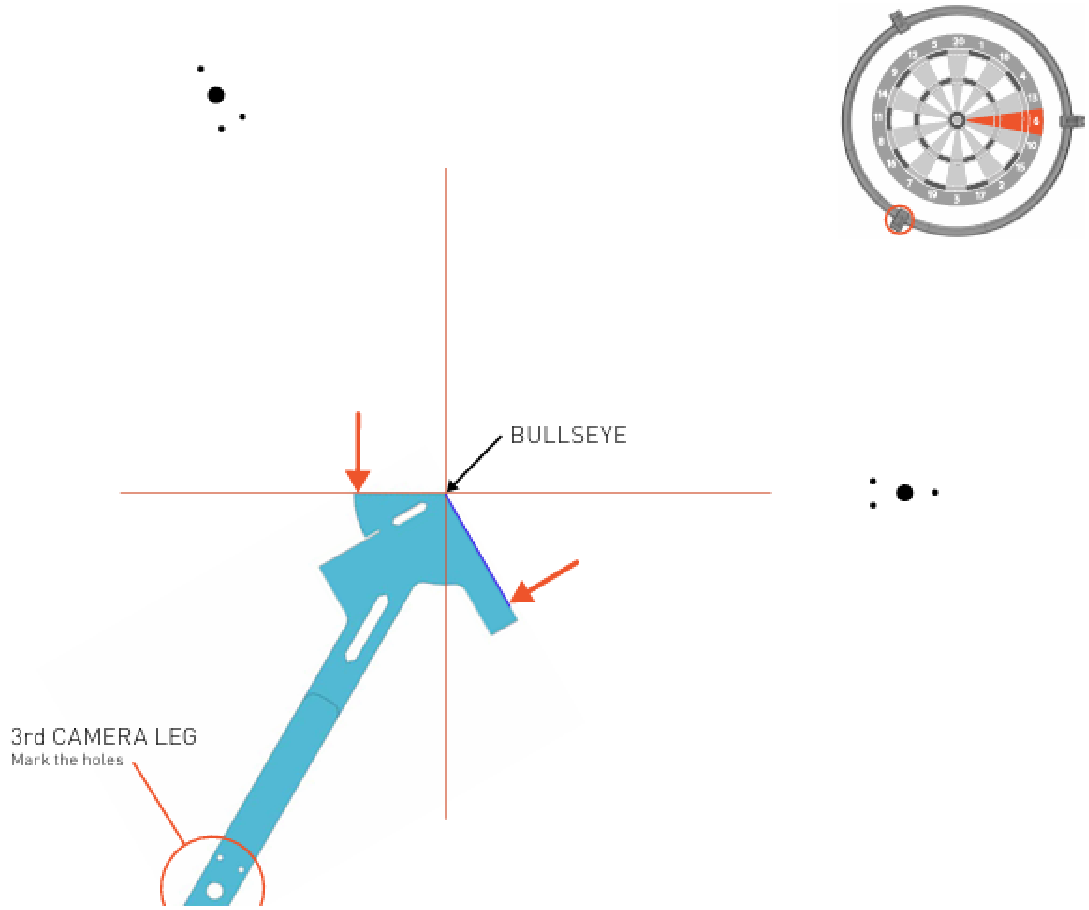
Mark the location of the second camera leg and draw a blue line.



Step 2 — Mark 2nd leg and draw blue line

Rotate the template clockwise again. Align it with the centre and the newly drawn blue line. Check the horizontal line again.

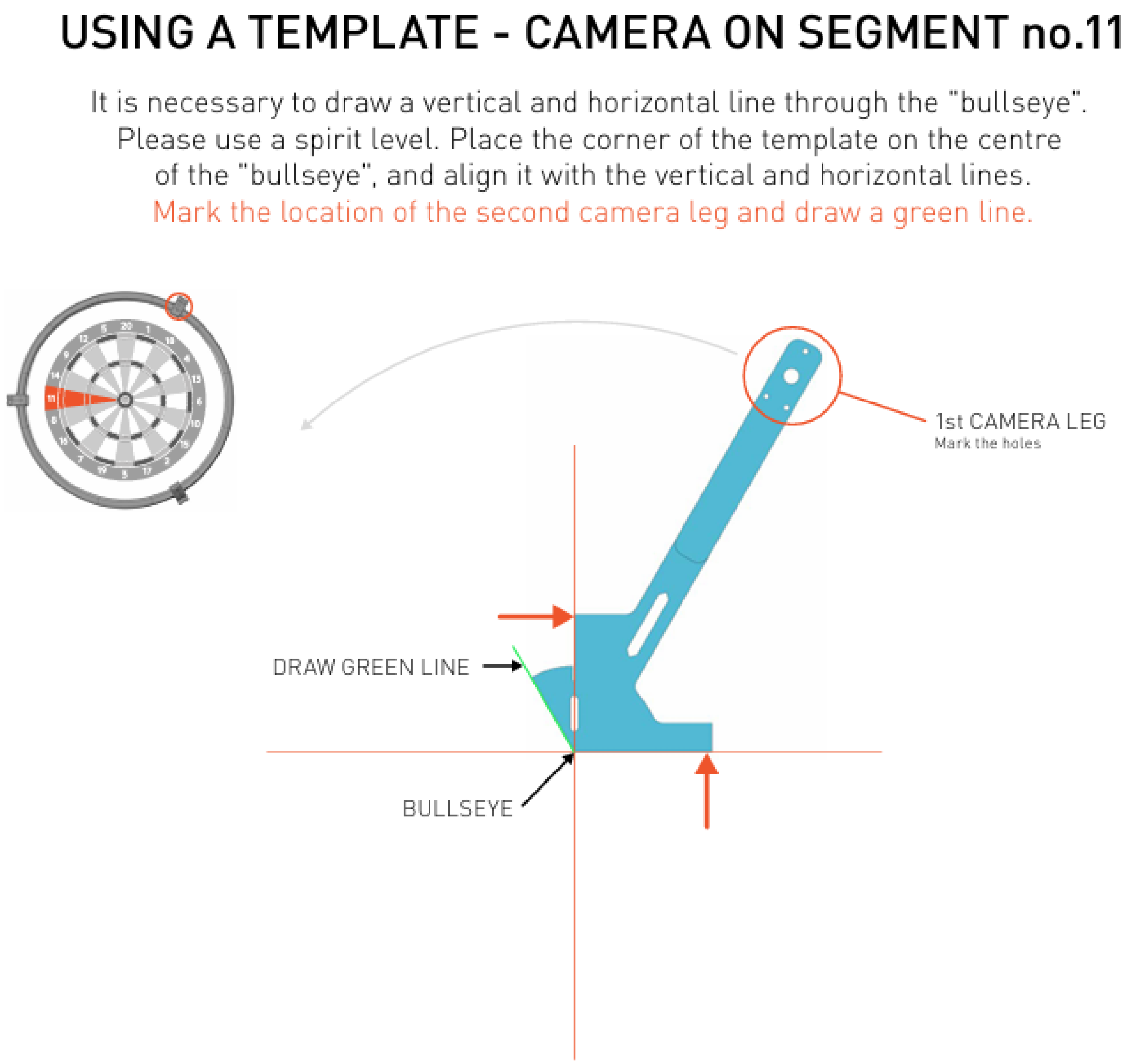
Mark the location of the third camera leg.



Step 3 — Mark 3rd leg position

Option B — Camera on Segment 11

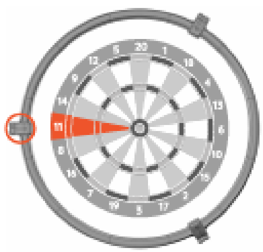
Step 1: Same starting position as Option A. Mark the 1st camera leg holes and draw a green reference line.

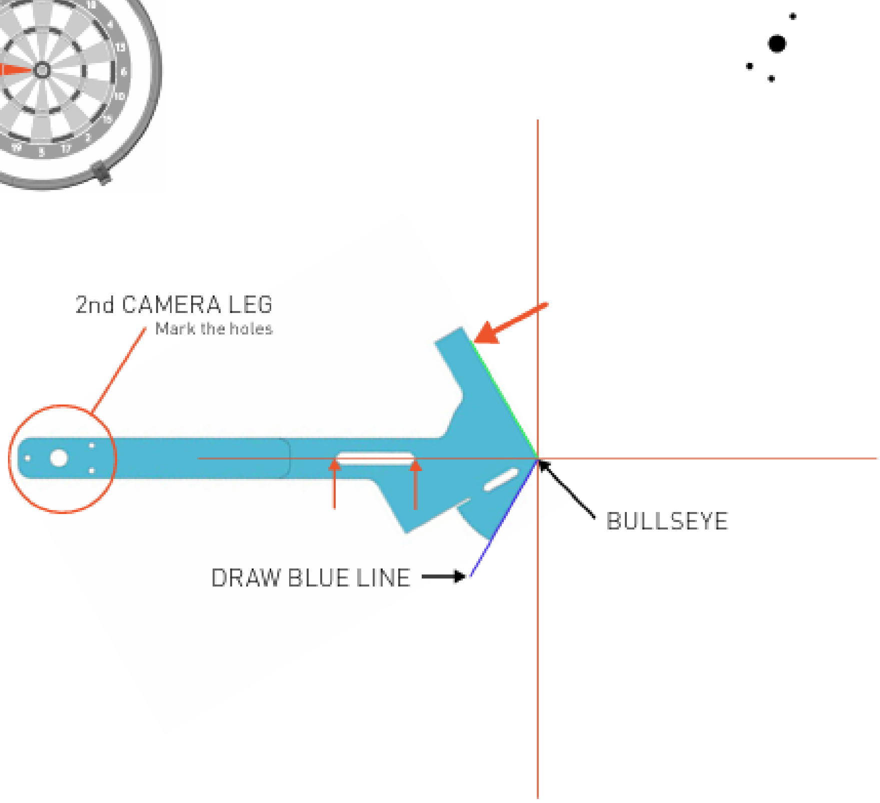


Step 1 — Mark 1st leg and draw green line

Rotate the model counuter- clockwise, and align it to the centre and the newly drawn green line. You can also check the horizontal direction through the template hole, the horizontal line should pass through the corners of the hole.

Mark the location of the second camera leg and draw a blue line.

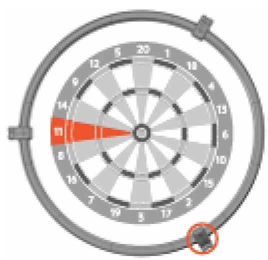


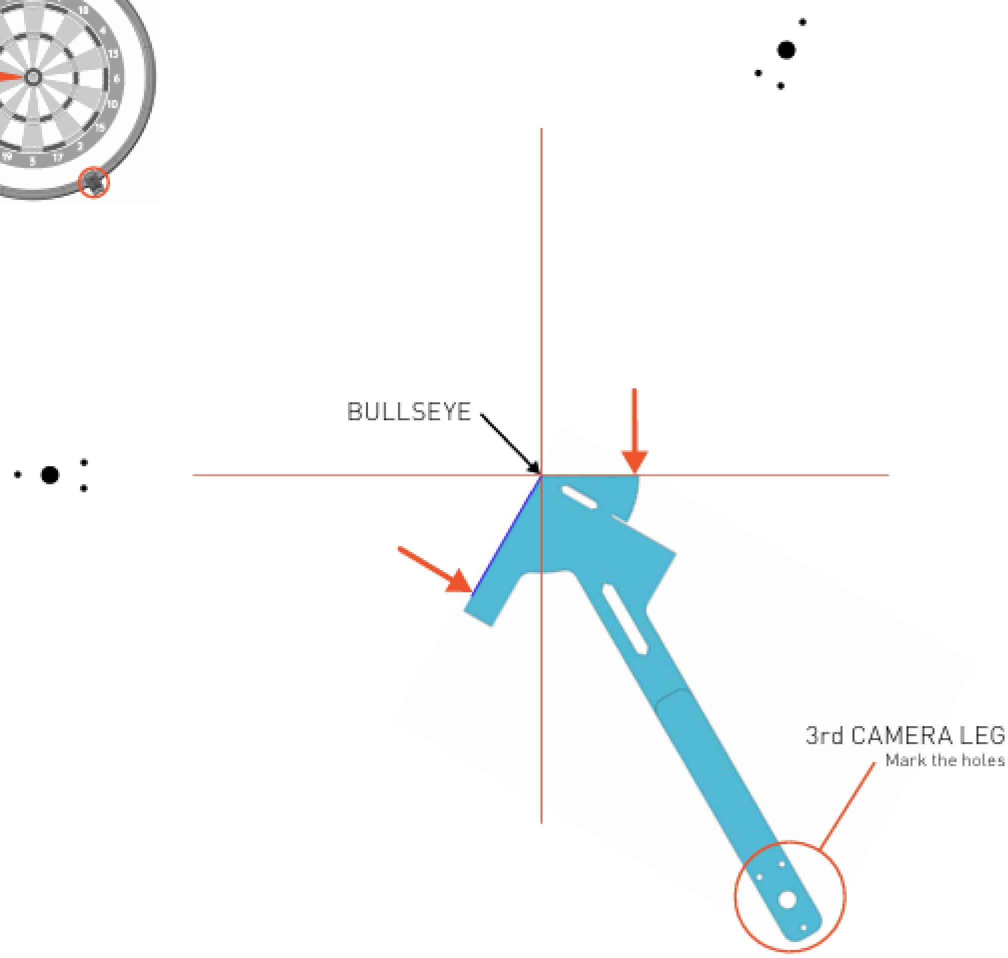


Step 2 — Mark 2nd leg and draw blue line

Rotate the template counter- clockwise again.
Align it with the centre and the newly drawn blue line.
Check the horizontal line again.

Mark the location of the third camera leg.



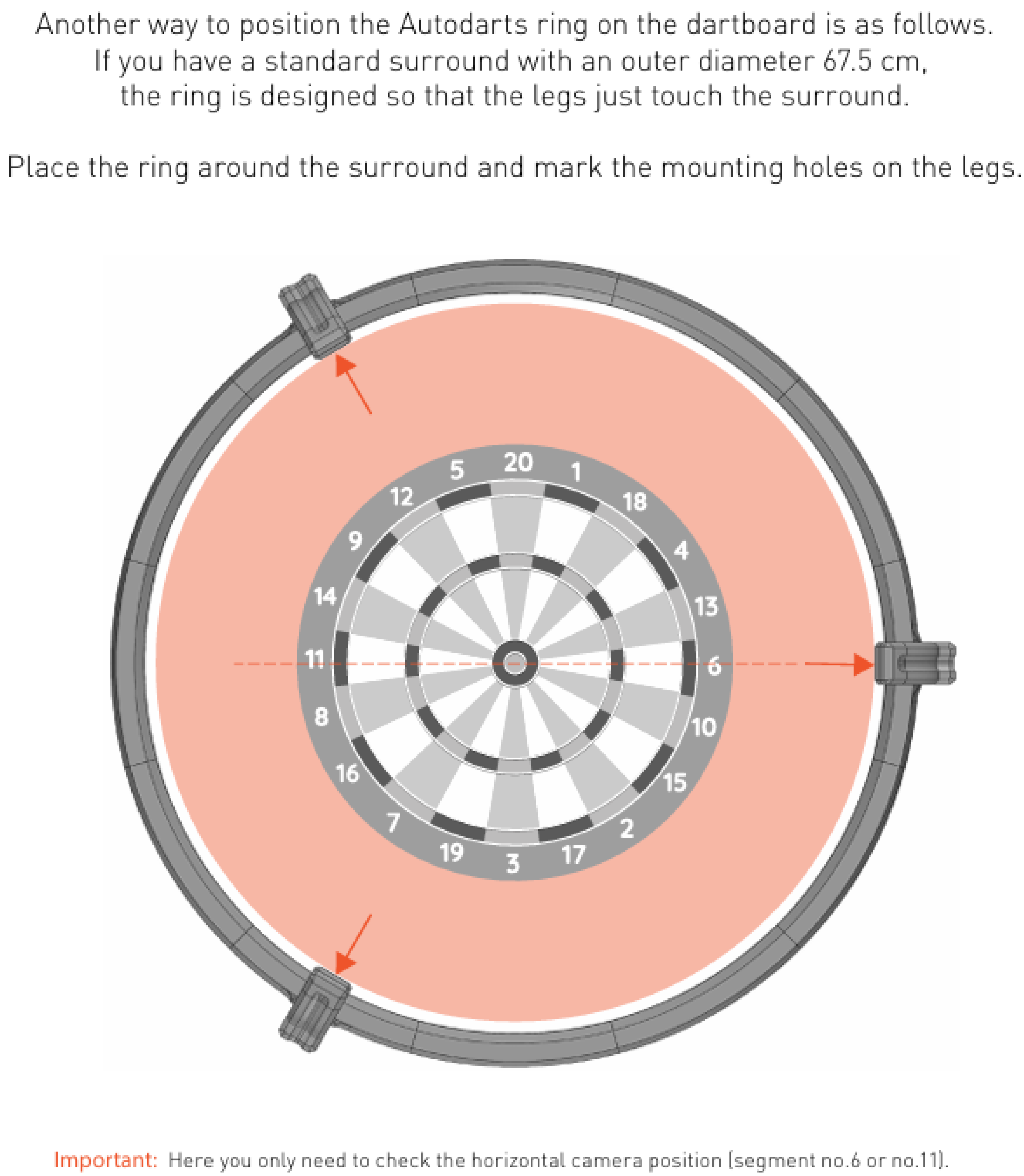


Step 3 — Mark 3rd leg position

2.6 Alternative: Position Using Surround

If you have a standard surround with an outer diameter of 67.5 cm, the ring legs are designed to just touch it. Simply place the ring around the surround and mark the mounting holes.

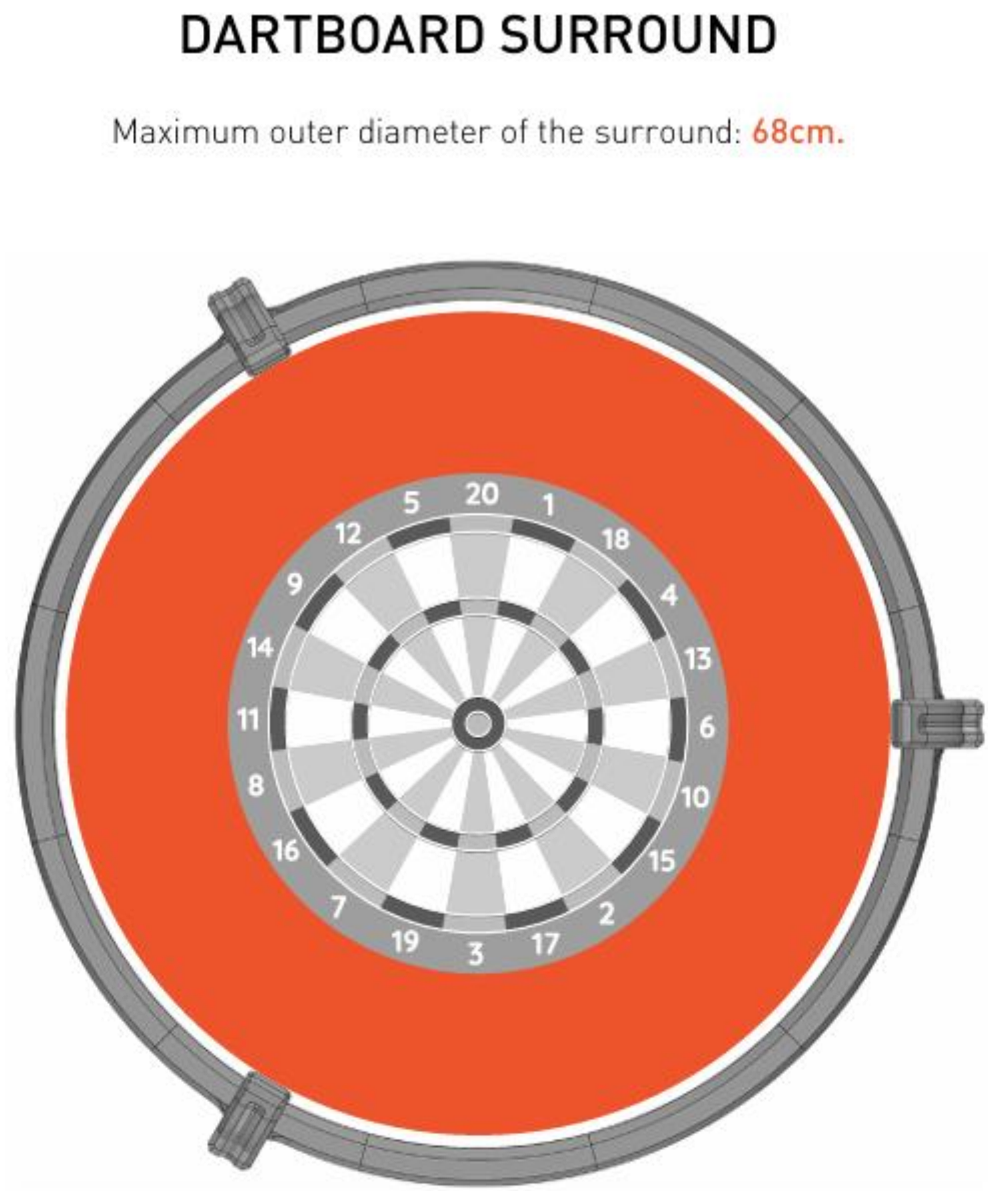
> ⚠️ *Important: You still need to confirm the horizontal camera position aligns with segment 6 or 11.*



Positioning the ring using a standard 67.5 cm surround

2.7 Dartboard Surround Clearance

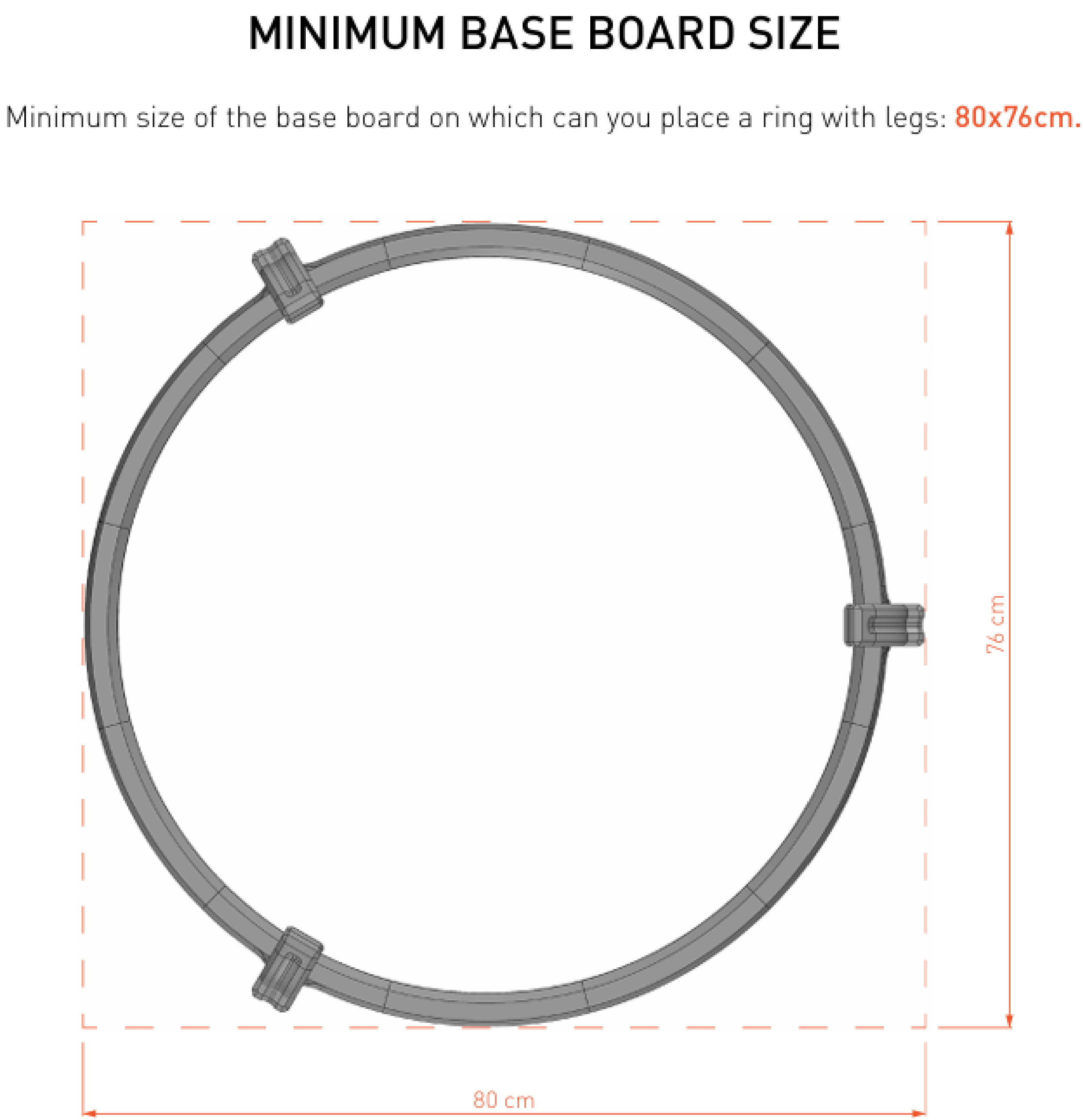
The maximum outer diameter of any dartboard surround used with this ring is 68 cm.



Maximum surround outer diameter: 68 cm

2.8 Minimum Backing Board Size

If mounting on a backing board, the minimum size to accommodate the ring with legs is 80 x 76 cm.

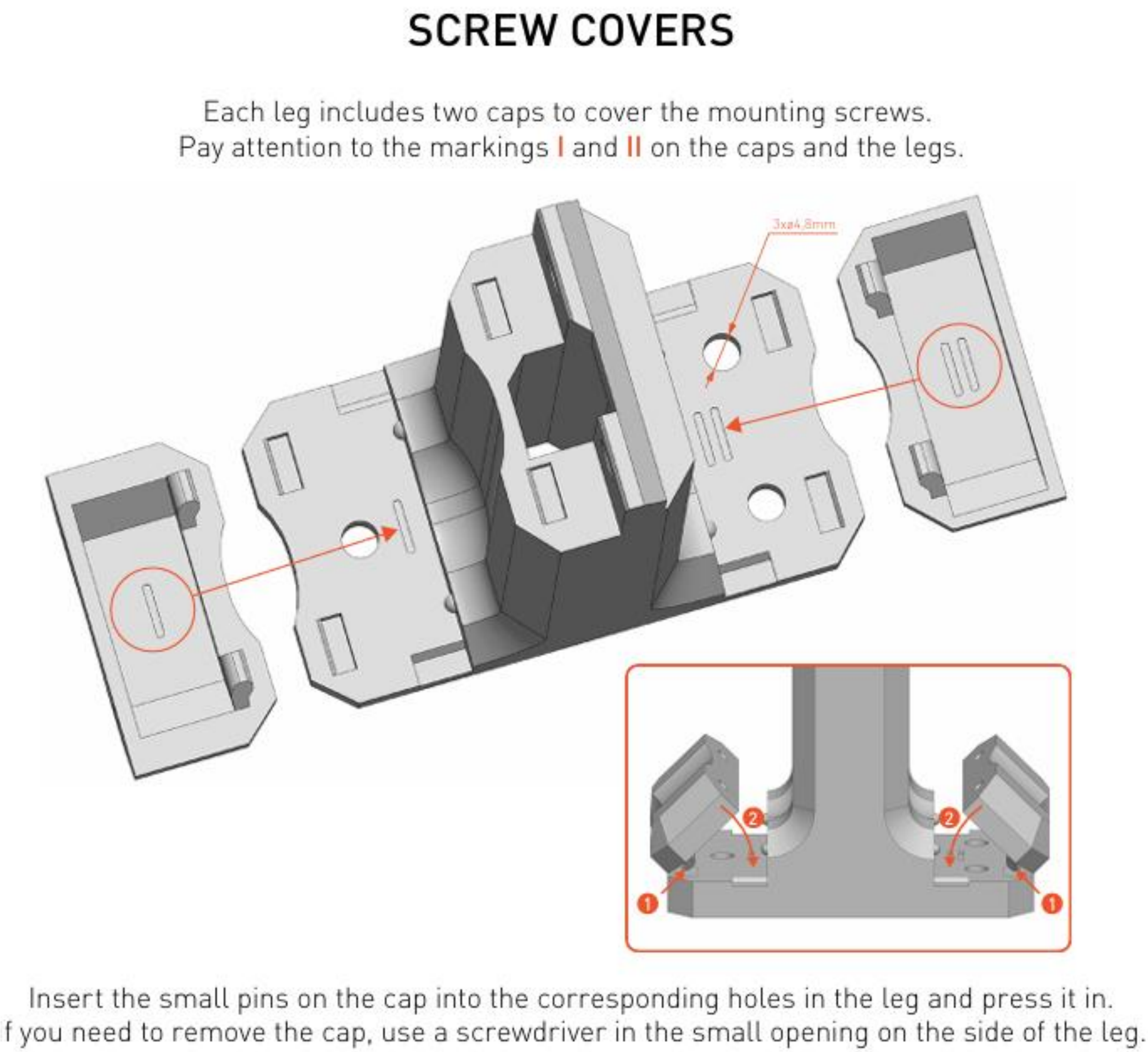


Minimum base board size: 80 x 76 cm

2.9 Screw Covers

Each leg includes two snap-on caps (marked I and II) to cover the mounting screws. Match the markings on the caps to the leg, insert the pins into the holes, and press in. Use a screwdriver in the side slot to remove them.

> ⚠️ *Use screws with a half-round head, max thread diameter ø4.5 mm. Oversized heads will prevent the caps from seating properly.*



Screw cover caps — note markings I and II must match the leg

3 PC Setup

Your DartCraft kit comes with a micro PC pre-configured with Ubuntu 24.04 LTS (XFCE desktop). It is set to automatically log in and launch the Autodarts Desktop app on startup — no manual configuration required.

🔐 **PC Login Credentials**

Username: autodarts **Password:** darts1234

Keep these safe — you may need them for future software updates.

3.1 Connections

Connect the following cables before powering on:

1. Plug each camera USB cable into a separate USB port on the PC. Do not use a USB hub — each camera must have a direct connection.
2. Connect the DisplayPort cable from the PC to your monitor or TV.
3. Plug the LED strip power adapter into a mains socket.
4. Connect the PC power cable and plug it into a mains socket.

3.2 Power On

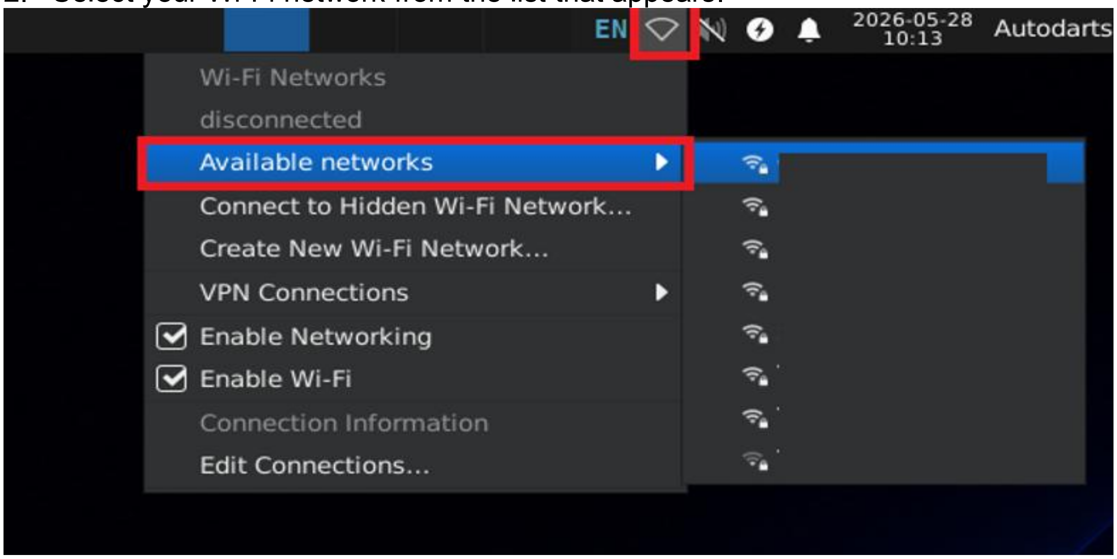
1. Press the power button on the front of the PC.
2. The PC will boot, automatically log in, and launch the Autodarts Desktop app. This takes approximately 30–60 seconds.

3.3 Connect to Wi-Fi (First Boot Only)

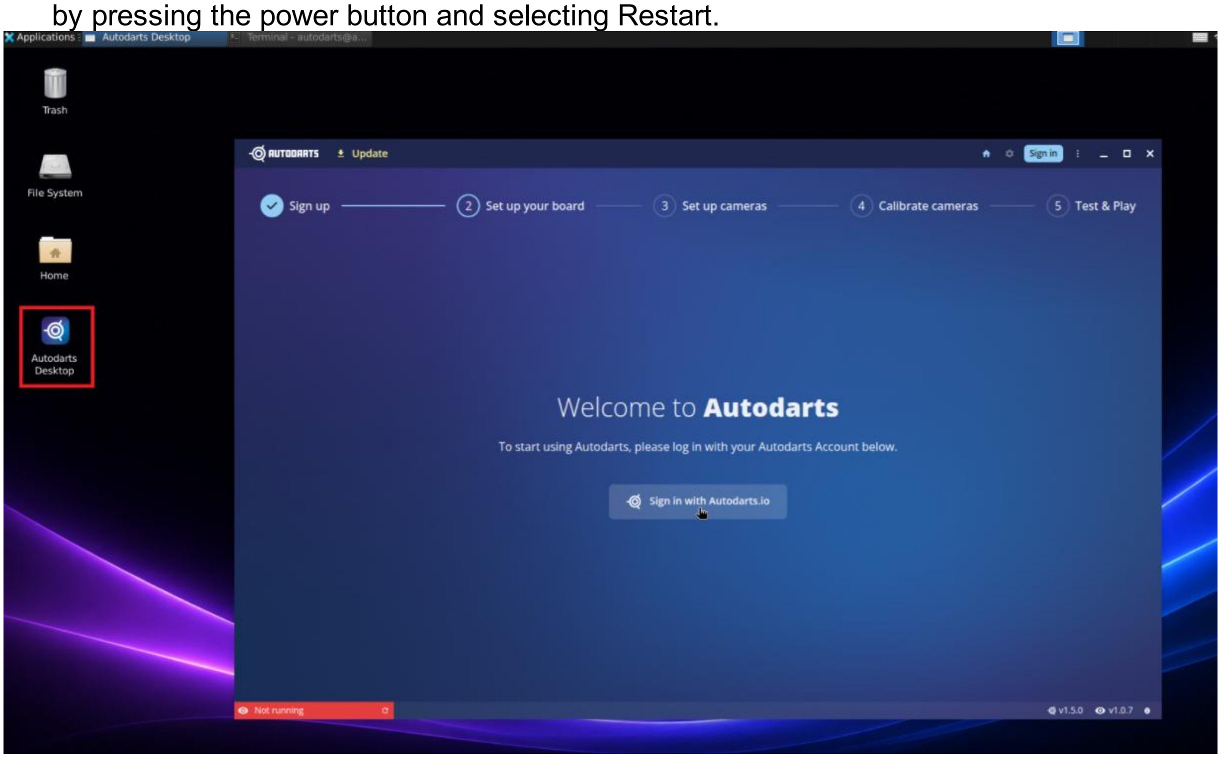
🔗 **Using a wired connection?:** If you have connected an Ethernet cable, you can skip this section — the PC will connect to the internet automatically.

To connect to your Wi-Fi network:

1. Click the network icon in the system tray at the bottom-right corner of the taskbar.
2. Select your Wi-Fi network from the list that appears.



3. Enter your Wi-Fi password and click Connect.
4. Once connected, double-click the Autodarts Desktop app. If it fails to open, reboot the computer by pressing the power button and selecting Restart.



3.4 Shutting Down

Always shut the system down properly before removing power from the PC.

Option A — Using the screen:

1. Click "Autodarts" in the top-right corner of the screen.
2. Select "Shut Down" from the menu.
3. In the confirmation dialog, click "Shut Down".
4. Once the PC has fully powered off, turn off the power to the LED ring.

Option B — Using the power button:

1. Press the power button on the front of the PC.
2. Select "Shut Down" when the dialog appears.
3. Once the PC has fully powered off, turn off the power to the LED ring.

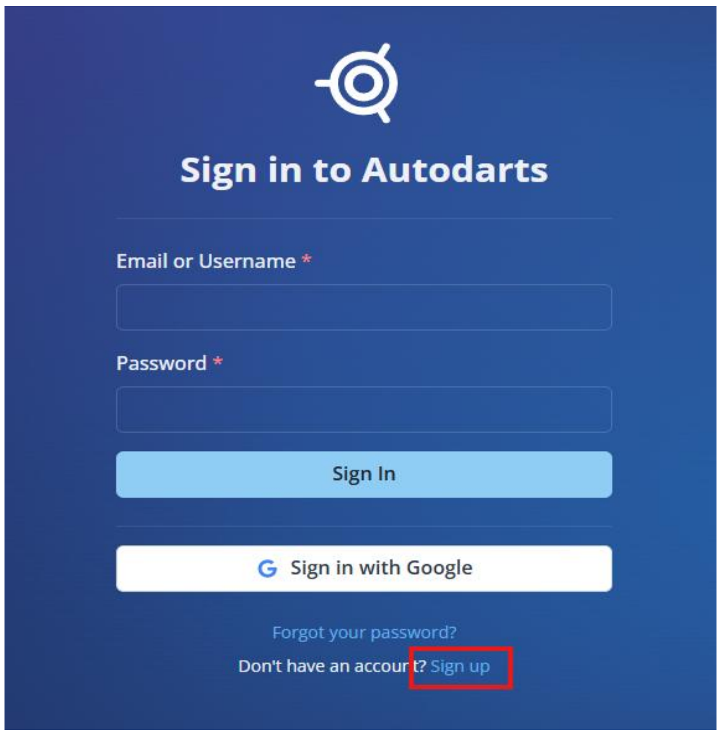
> ⚠️ **Important:** Never unplug the PC from the wall while it is running. Always shut it down first to avoid data loss or system errors.

4 Autodarts Setup

Autodarts is the software platform used to detect dart throws and manage your games. Follow the steps below to create your account, link your board, and calibrate the cameras.

4.1 Create an Autodarts Account

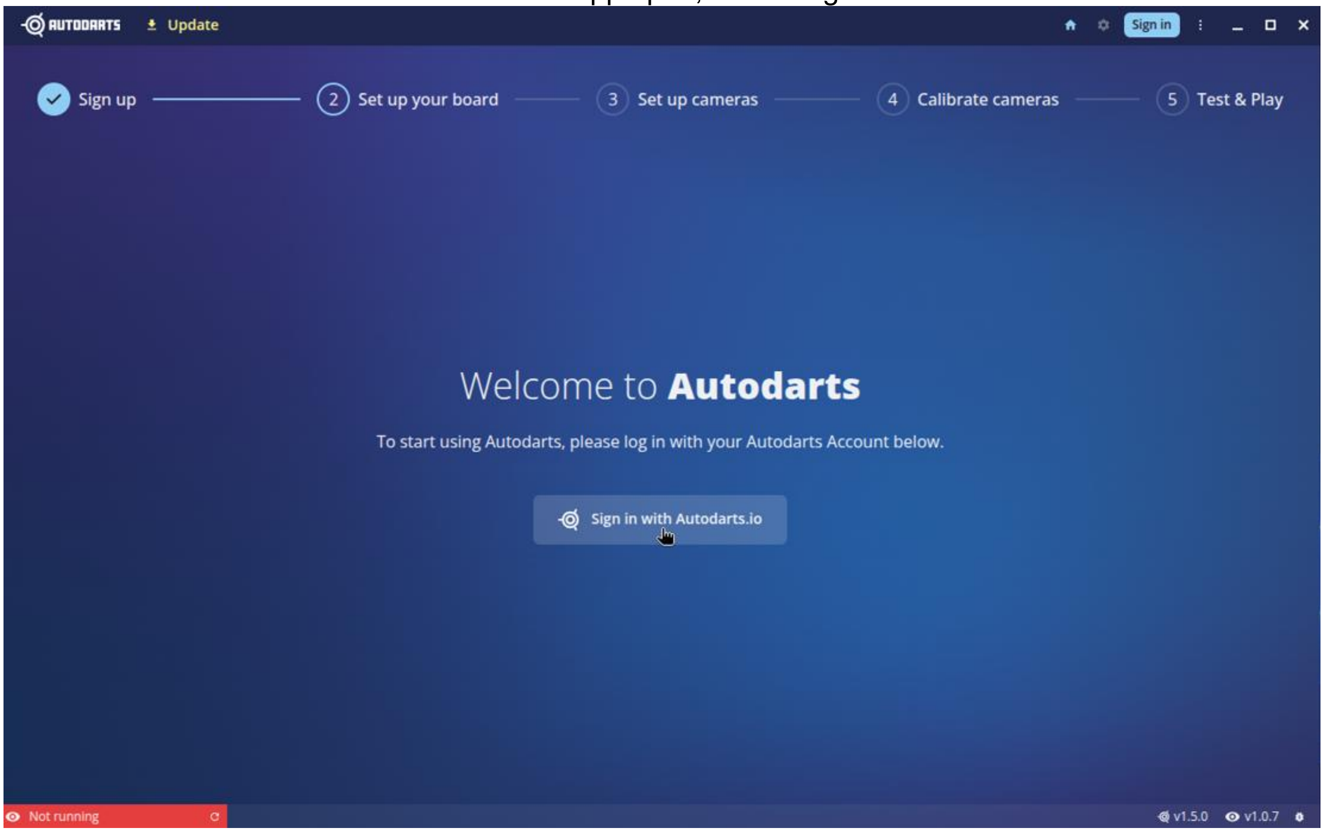
1. Either on the PC or your phone navigate to https://play.autodarts.io and select Sign Up



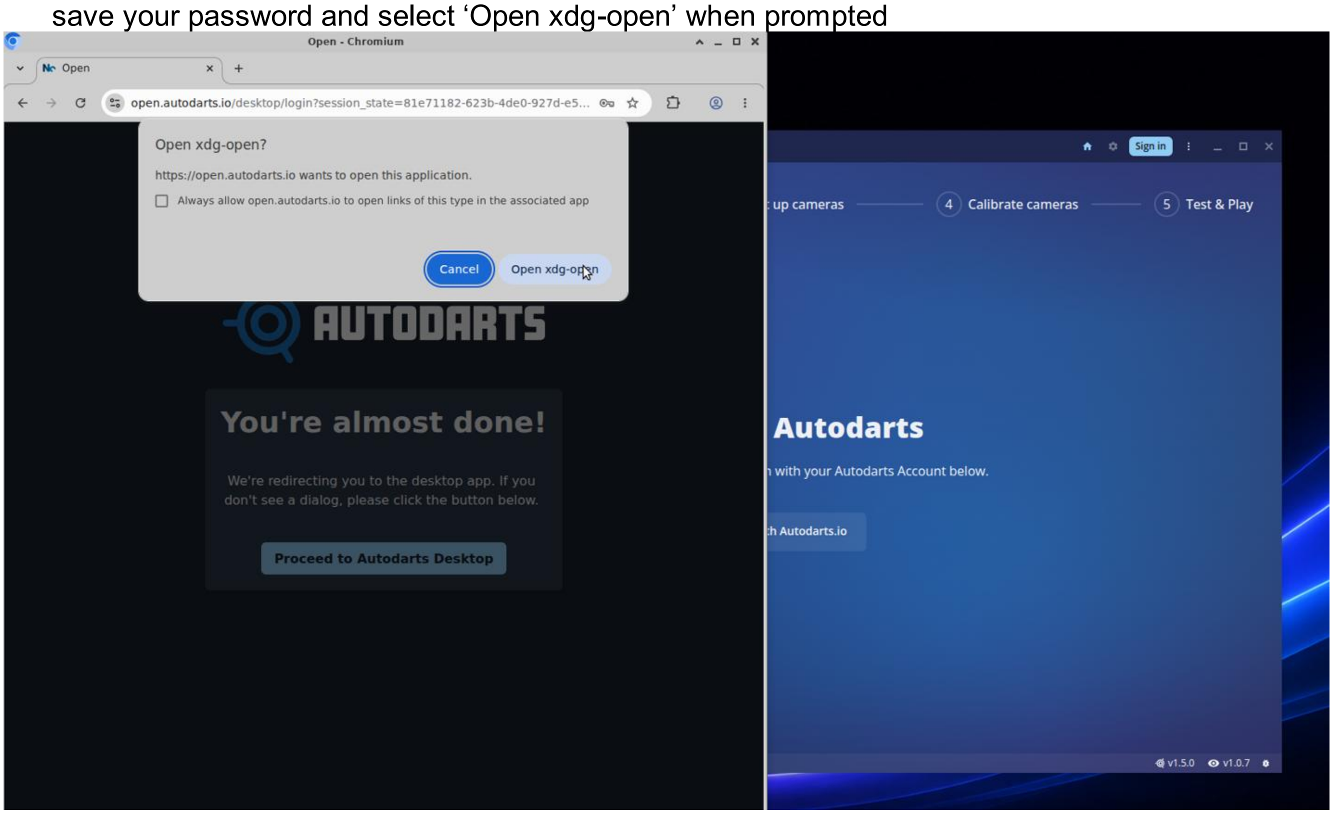
2. Fill in the required details and select Sign Up.
3. Check your email and click the verification link.

4.2 Sign In & Link Your Board

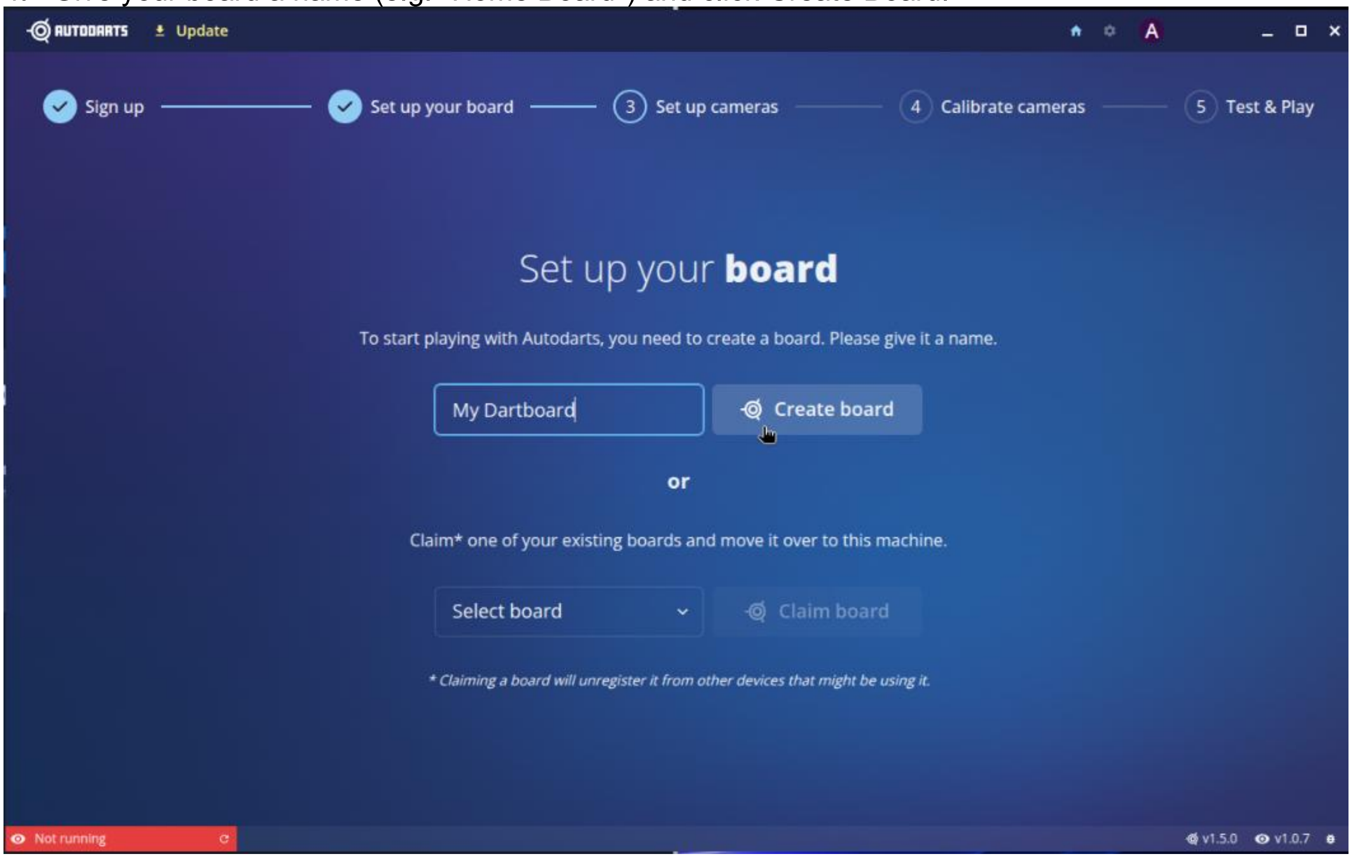
1. On the Autodarts PC with the Autodarts app open, select Sign In.



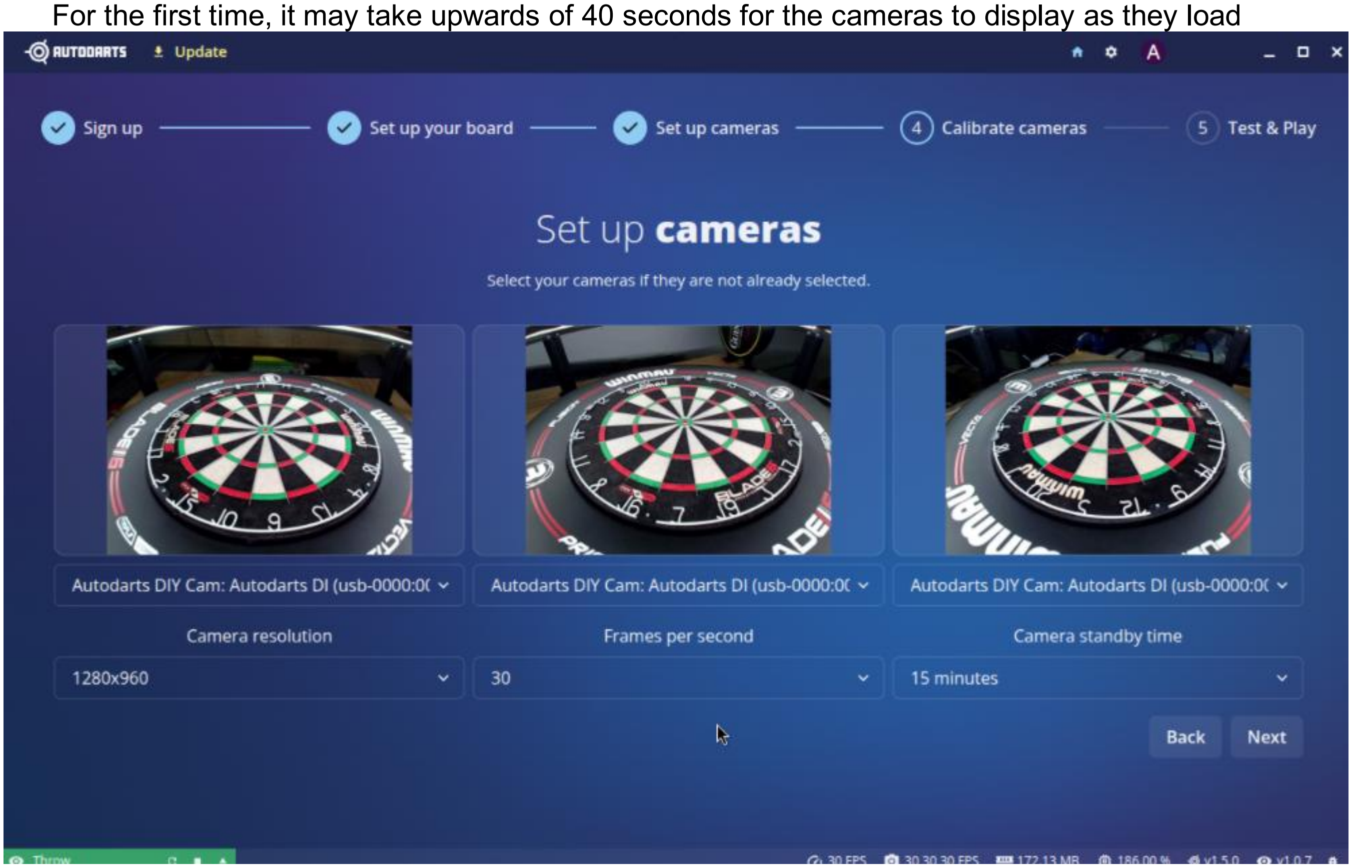
2. A web browser window will open for you to sign in with your Autodarts account. Once signed in, save your password and select ‘Open xdg-open’ when prompted



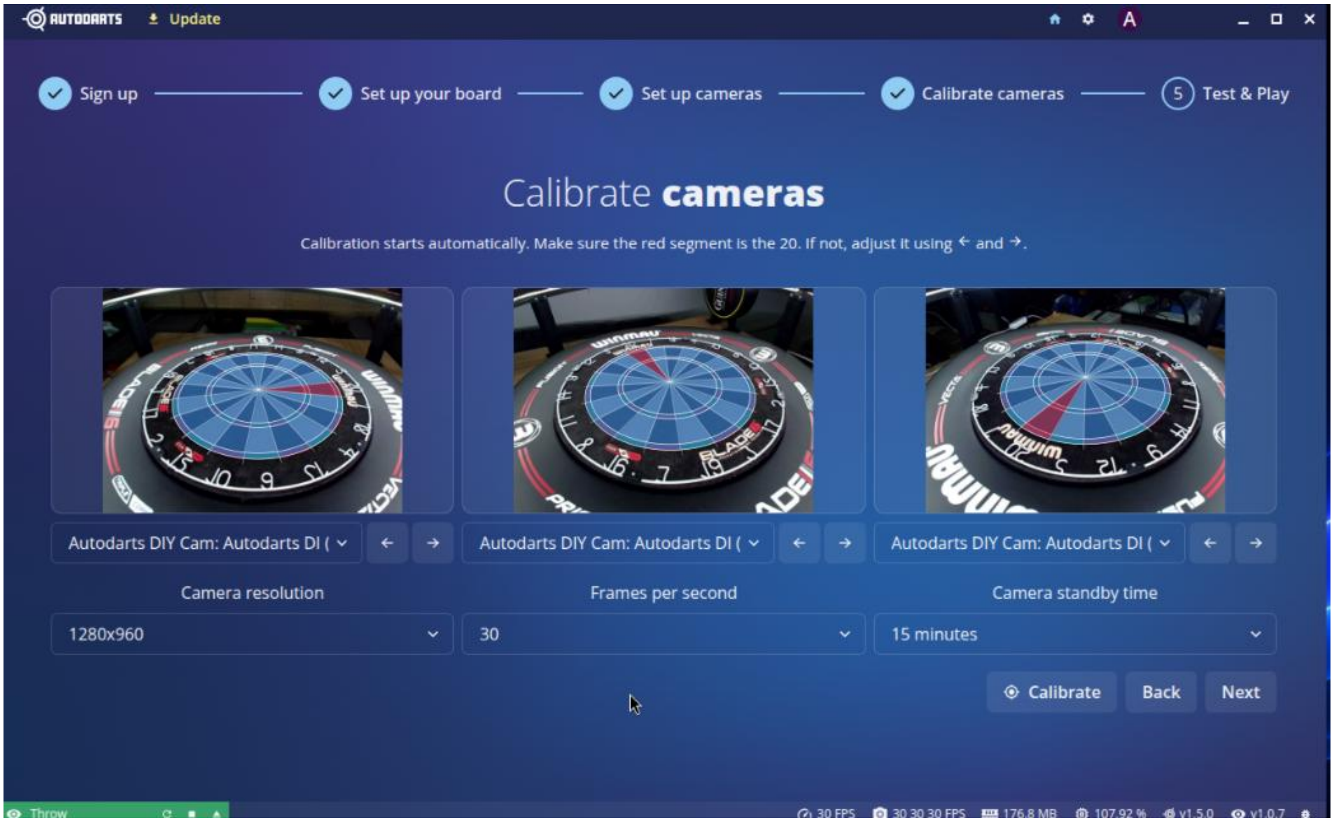
3. The Autodarts app will sign into your account within 30 seconds, ready to set up your board
4. Give your board a name (e.g. "Home Board") and click Create Board.



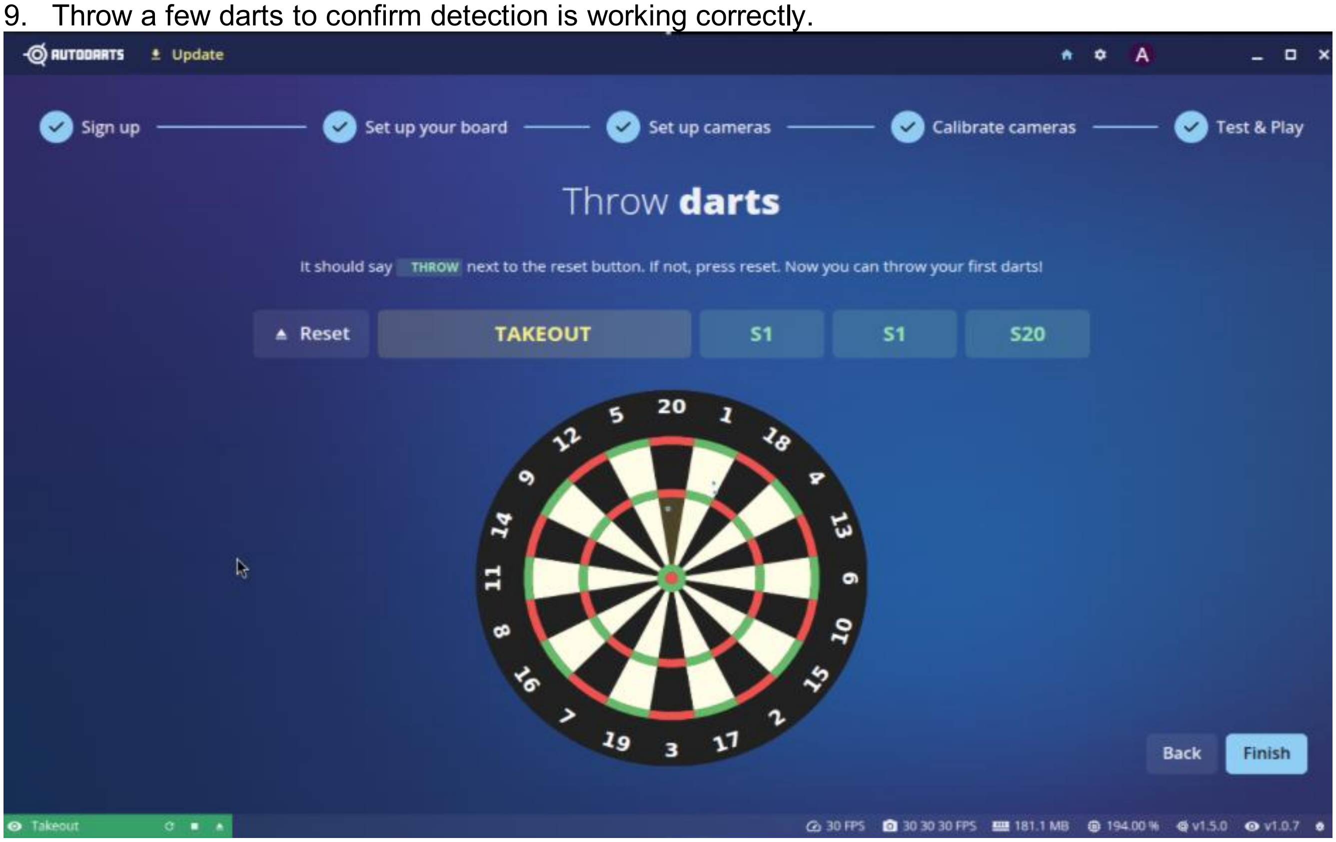
5. You will see a live image from each of the three cameras. Verify all three are showing clearly. For the first time, it may take upwards of 40 seconds for the cameras to display as they load



6. Click Next.
7. The system will automatically calibrate and you should now see the detection overlay showing the different sections of the board.



8. Click Next.
9. Throw a few darts to confirm detection is working correctly.



10. Click Finish.

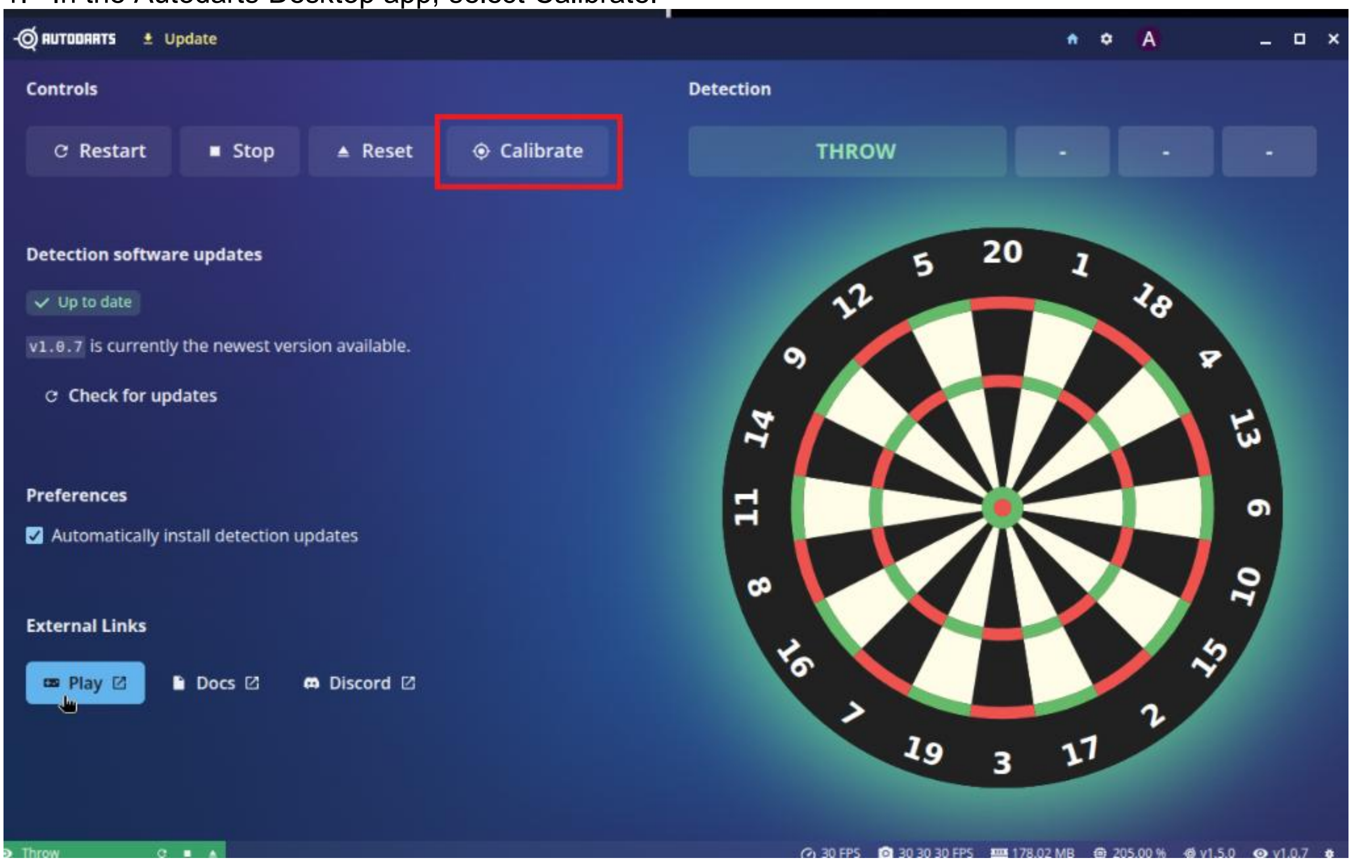
4.3 How to Calibrate the System

Calibration tells the software exactly where the scoring segments are. You should recalibrate whenever:

- The board or cameras have been moved or knocked
- Detection accuracy seems reduced
- You have replaced or repositioned the LED strip

To calibrate:

1. In the Autodarts Desktop app, select Calibrate.



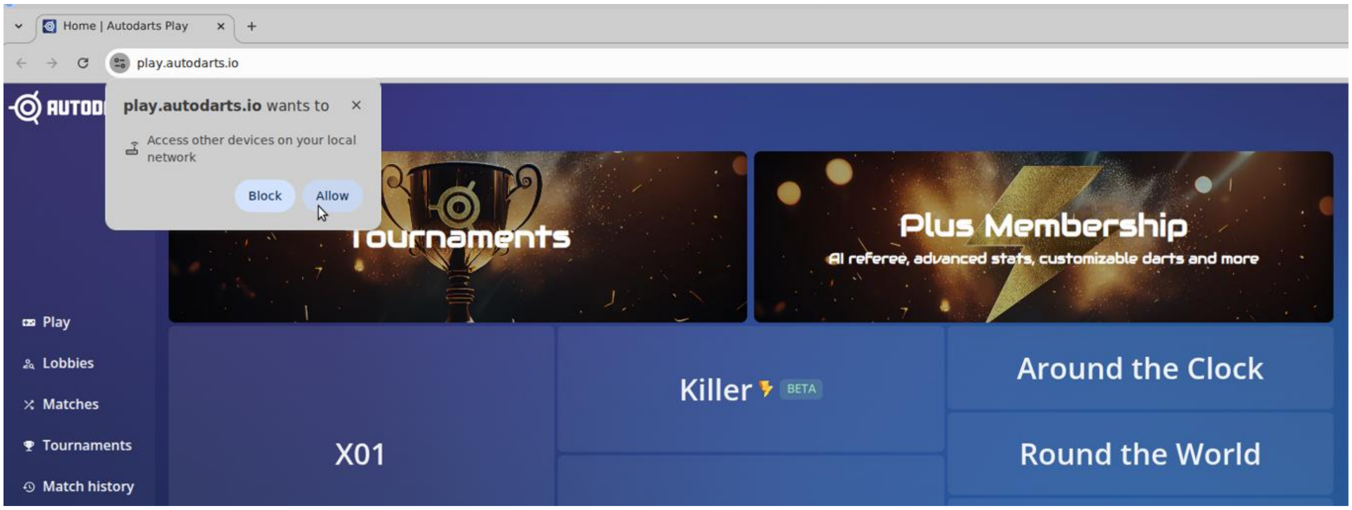
2. The glow around the board will turn purple during calibration and return to green once complete.

Important — before calibrating: Make sure the LED strip is switched on, the dartboard surface is clean, and no objects are obstructing the camera view.

5 Starting a Game

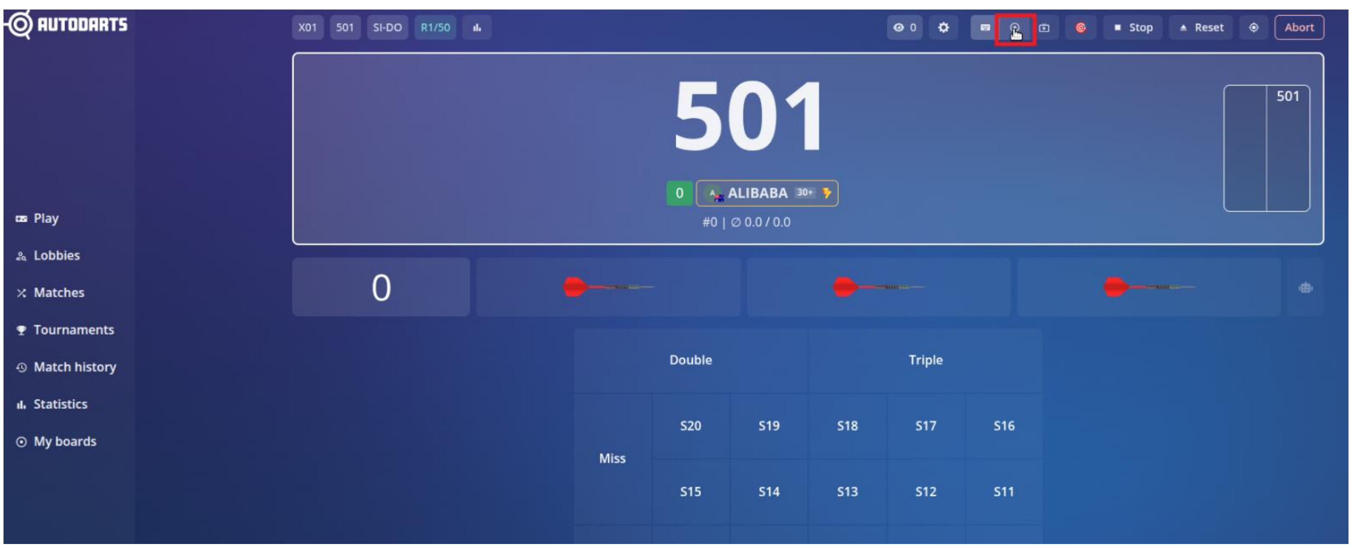
Once calibrated and connected, you're ready to play. Click Play in the AutoDarts app — a browser window will open and log you in automatically if you've saved your credentials, or sign in and save them for next time.

If prompted for the first time, select allow



1. Select a game mode (e.g. 501, Cricket, Training games).
2. Configure game settings to your preferences, then select Open Lobby.
3. Add local players, bots, or invite online friends.
4. Click Start Game when ready.

Tip: When in a game you can select this button at the top to show a board view instead.



💡 *Set your game lobby to Private if you do not want random players joining online.*

6 Troubleshooting

Issue	What to try
Throws not registering	Check that cameras are not blocked and that lighting is consistent.
Inaccurate scoring	Re-run calibration. Ensure the board is level and nothing has shifted.
Camera not detected	Check USB cables are firmly connected to the PC. Try a different USB port. Restart the Autodarts app.
Wi-Fi not connecting	Restart the system. Move closer to your router or use an Ethernet cable.
Camera orientation wrong	In Autodarts Board Manager, go to Camera Settings and use the Rotate option to correct each camera view.
Calibration fails	Ensure lighting is consistent, cameras are not obstructed, and the ring is centred on the board. Check that all cameras can clearly see the full board, then try again.
Auto calibration keeps failing	Try manual calibration if available. Auto calibration can be affected by lighting, camera angles, reflections, or board movement.
Only one or two cameras showing	Check each camera cable individually. Try swapping USB ports. If using a USB hub, make sure it is powered or try connecting cameras directly to the PC.
Camera feed is dark or blurry	Check the LED lighting is on and evenly lighting the board. Clean the camera lenses carefully if needed. Avoid shadows across the board.
Glare or reflections on the board	Adjust nearby lighting or reduce bright reflections from windows, lamps, or glossy surfaces. Consistent, even lighting gives the best results.
Throws appear in the wrong area	Check that each camera is assigned and oriented correctly, and that the board has not rotated or shifted after calibration. Re-run calibration after making any adjustments.
Scoring was accurate, then became inaccurate	Check whether the dartboard, surround, or ring has moved. Even small movement can affect accuracy. Re-centre the ring and recalibrate.
Lag or slow response	Restart the PC and close any other open applications. If using Wi-Fi, try switching to Ethernet. On the XFCE desktop, right-click the taskbar and open Task Manager to check for high CPU or memory usage.
USB devices disconnect randomly	Try a different USB port or use a powered USB hub. Avoid loose connections or placing tension on the camera cables.

Keyboard or mouse not responding	Check the USB receiver is plugged in. Replace or recharge the keyboard batteries if needed. Try moving the receiver to a different USB port.
Sound not working	Check the speaker is powered on, the USB power cable is connected, and the 3.5mm jack is firmly in the headphone port. On the XFCE desktop, right-click the volume icon in the taskbar and confirm the correct audio output device is selected.

💡 **Tip:** If you are unsure what the cameras are seeing, open Autodarts Board Manager and check the live camera view. This is the quickest way to spot blocked cameras, poor lighting, incorrect rotation, or alignment issues.

⚠️ **After any movement:** If the dartboard, surround, ring, or cameras are moved for any reason, calibration should be checked and re-run before playing again.

7 Support

DartCraft kits are sold exclusively through dartcraft.com.au. If you experience a hardware issue — such as a faulty camera or damaged component — please get in touch and we'll make it right.

Hardware Support	Email **hello@dartcraft.com.au** with as much detail as possible, including photos or images if you can.
Autodarts Community	For software questions, calibration tips, and general help, join the Autodarts Discord: **discord.com/invite/autodarts**
Documentation	Full Autodarts documentation is available at: **autodarts.diy**

DartCraft is an independent hardware seller and is not affiliated with, endorsed by, or officially supported by Autodarts.

Thank you for choosing **DartCraft** — enjoy your game!